

Realization-Dependent Microstructure and Statistical Morphology Emulation in Ballistic Helical GLAD

Oussama Ben Haj Ali^{1*}, Ferid Chaffar Akkari¹, Bruno Gallas²,
Nicolas Martin³

^{1*}Laboratoire de Photovoltaïque et Matériaux Semi-conducteurs
(LPMS), École Nationale d'Ingénieurs de Tunis (ENIT), Université de
Tunis El Manar (UTM), Tunis, Tunisia.

²Institut des NanoSciences de Paris (INSP), CNRS, Sorbonne
Université, Paris, France.

³Institut FEMTO-ST, CNRS, Université de Franche-Comté, Besançon,
France.

*Corresponding author(s). E-mail(s): oussama.benhadjali@fst.utm.tn;

Abstract

Glancing-angle deposition (GLAD) links a single controlled parameter, the deposition angle α , to film porosity and to the optical anisotropy relevant to gas-sensor and polarimetric-device performance, but the range of structural information a ballistic simulator can predict from α has not been systematically established. We characterise a bead-sphere ballistic GLAD simulator across twelve helical Cu configurations ($\alpha = 60^\circ\text{--}89^\circ$, $h = 300$ nm, pitch = 150 nm). Void fraction increases monotonically from **88.1%** to **98.5%** with sub-**0.25%** seed-to-seed variability at this box/height convention (degrading at larger lateral box size; Appendix A). A corotating-frame oracle test shows absolute column positions carry no realization-invariant signal recoverable by a coordinate-conditioned model, establishing this boundary of predictive reach as a central, deliberately reported result. By contrast, the two-point pair correlation $h_g(\mathbf{r}; \alpha)$ is reproducible and varies smoothly with α : a compact emulator (CORRNIF) reproduces held-out angles with interior leave-one-angle-out $R^2 = \mathbf{0.992}$, though not more accurately than simple linear or PCHIP interpolation on the same folds, and reproduces a genuinely

unseen prospective angle ($\alpha = 72^\circ$, confirmed by a dedicated simulation) with $R^2 = 0.9907$, again no better than linear interpolation ($R^2 = 0.9909$). CORRNIF's only demonstrated advantage over these simpler baselines is exercised directly: gradient descent through the frozen, trained model recovers the deposition angle behind a held-out target curve from five different starting points to the same value, matching an independent grid search to within 0.002° . An illustrative aligned-cylinder Maxwell-Garnett calculation for tungsten columns gives form birefringence $\Delta n \sim 0.3\text{--}0.8$, but strong extraordinary absorption and sensitivity to the filling-fraction definition mean no polarimetric design window can be claimed; this is reported as a loss-aware screening illustration, not a device prediction. The result is a simulation-only workflow for ranking helical GLAD geometries by ensemble morphology, with the limits of predictive reach established as an explicit finding and quantitative optical/material calibration left to dedicated experiments.

Keywords: glancing-angle deposition; sculptured thin films; columnar nanostructures; bead-sphere ballistic simulation; realization dependence; void fraction; pair correlation function; statistical morphology emulation; effective-medium screening; neural implicit field

1 Introduction

Glancing-angle deposition (GLAD) exploits extreme substrate inclination ($\alpha > 70^\circ$) to produce self-assembled nanocolumnar thin films with tunable porosity and helical geometry, an approach whose sculptured-column control originates with the chiral thin films of Robbie et al. [1] and has since been reviewed extensively [see reviews by 2–6]. The physical origin and evolution of the columnar, shadowing-driven morphology itself has been characterised in detail [7, 8]. The helical architecture generates strong geometric form birefringence: the ordinary and extraordinary refractive indices of the nanostructured film differ because oriented columnar inclusions interact differently with polarisation states parallel and perpendicular to the helix axis [9]. This tuneable optical anisotropy underlies the application of GLAD films in ellipsometric waveplates, chiral-selective filters, and label-free polarimetric biosensors where large phase retardation between the two polarisation eigenstates is required [10, 11]. Recent

demonstrations of Cu₂O GLAD films as gas-sensing platforms underline the practical urgency of predictive design tools: surface roughness and optical anisotropy both depend sensitively on α , and model-guided angle selection is critical for optimising sensor response [12]. Realising this design potential demands models that reliably link α to the structural filling fraction $f(\alpha)$ and thence to effective optical anisotropy.

Ballistic simulation models, which track straight-line atom trajectories without thermally activated surface diffusion, are the fastest tool for screening deposition-angle effects on void fraction and columnar geometry at the nanometre scale [13]. The self-shadowing tangent rule underlying such models traces back to the microfractographic columnar-tilt observations of Nieuwenhuizen and Haanstra [14], was formalised into predictive columnar-growth models by Tait et al. [15] and Lichter and Chen [16], and was extended to full three-dimensional ballistic simulation—of which the present simulator is a direct descendant—by Smy et al. [17]; the effective surface area generated by such oblique shadowing growth has been analysed numerically by Suzuki and Taga [18]. A more recent, independently implemented ray-traced Monte-Carlo 3D GLAD simulator, validated against the same column-growth-angle relation, is given by Kulkarni et al. [19], illustrating that this modelling family continues to be actively re-implemented rather than being a single inherited codebase. Despite their computational efficiency, ballistic models are known to over-predict porosity relative to experimentally deposited films because adatom surface diffusion tends to densify the structure after impact [20, 21]. The recent success of neural implicit fields (NIFs)—compact multi-layer perceptrons that map continuous coordinates to scalar field values, as established for shape and scene representation by Sitzmann et al. [22], Park et al. [23], and Mescheder et al. [24], and made practical at high spatial resolution via multiresolution hash encodings [25]—in representing three-dimensional structures has raised the question of whether such models can learn and interpolate

139 GLAD column morphology directly from simulation data, enabling rapid parameter-
 140 space exploration without re-running the simulator at every deposition angle [26].
 141 However, neither the fundamental predictability of per-voxel column positions nor
 142 the reproducibility and interpolability of statistical descriptors extracted from GLAD
 143 simulation data has been systematically established.
 144
 145 Here we address this gap with a combined simulation-characterisation and
 146 machine-learning study of the bead-sphere ballistic GLAD simulator across twelve heli-
 147 cal Cu configurations spanning $\alpha = 60^\circ\text{--}89^\circ$ ($h = 300\text{ nm}$, pitch = 150 nm). Our cen-
 148 tral machine-learning contribution is a principled two-level investigation of data-driven
 149 morphology modelling. First, a corotating-frame oracle experiment shows empirically
 150 that absolute per-voxel column positions are not a target a coordinate-conditioned
 151 neural implicit field can recover under the multi-alpha stochastic-nucleation protocol
 152 of this simulator: the best fixed spatial template gives no gain over the bulk-filling-
 153 fraction baseline, because column positions are re-randomised in every independent
 154 run. Second, we demonstrate that the normalised two-point pair correlation $h_g(r; \alpha)$ —
 155 an approximately reproducible ensemble descriptor under the tested seeds—is instead
 156 smoothly interpolable: a compact parametric emulator (CORRNIF) trained on six
 157 deposition angles reproduces $h_g(r; \alpha)$ at unseen interior angles with mean leave-one-
 158 angle-out $R^2 = 0.992$. A prospective validation at $\alpha = 72^\circ$, genuinely absent from the
 159 training set and subsequently confirmed by a dedicated simulation, is reproduced by
 160 CORRNIF with $R^2 = 0.9907$, while a simple linear interpolation gives $R^2 = 0.9909$ on
 161 the same corrected target. We additionally quantify the cross-material, cross-method
 162 discrepancy between the ballistic model and published optical void fractions and, com-
 163 bining the extracted filling fractions $f(\alpha)$ with the aligned-cylinder Maxwell-Garnett
 164 effective-medium model [27], provide a loss-aware birefringence screening illustration
 165 for high-index columnar geometries.
 166
 167
 168
 169
 170
 171
 172
 173
 174
 175
 176
 177
 178
 179
 180
 181
 182
 183
 184

Concretely, this paper answers four questions that an experimentalist choosing a	185
deposition angle, or a modeller deciding what to simulate next, would need answered	186
before trusting a ballistic GLAD simulation:	187
	188
	189
Q1 How does bead-sphere void fraction depend on α in this ballistic model, how	190
reproducible is it across independent stochastic runs, and how does it compare	191
quantitatively with published optical void fractions of other GLAD metals?	192
	193
Q2 Does the film’s internal structure vary with depth in a way that matters for prop-	194
erty predictions, and can a small set of bulk morphology descriptors (openness,	195
anisotropy, width gradient) be summarised in a low-dimensional space across the	196
angle sweep?	197
	198
Q3 Can a machine-learning model (a neural implicit field) learn to predict GLAD mor-	199
phology from simulation data, and if so, at what level—individual column positions,	200
or ensemble statistics—is such a prediction actually possible and reproducible?	201
	202
Q4 What do the resulting morphology descriptors imply, even only illustratively, for the	203
optical anisotropy that end applications (polarimetric sensors, waveplates) depend	204
on?	205
	206
	207
The Conclusion (Section 6) closes against these four questions in the same order.	208
Specifically, we report: (i) the bead-sphere void-fraction trend and its three-seed	209
reproducibility (Section 3.1, Q1); (ii) depth-resolved void-fraction profiles that reveal	210
growth-front evolution (Section 3.2, Q2); (iii) a morphology-space analysis by principal	211
component analysis (Section 3.3, Q2); (iv) a quantitative cross-material, cross-method	212
discrepancy of +20.7 pp against published optical void fractions (Section 3.4, Q1);	213
and (v) a full analysis: a corotating-frame oracle bounding absolute-position recov-	214
ery, CORRNIF pair-correlation interpolation with interior leave-one-angle-out $R^2 =$	215
0.992, prospective validation at $\alpha = 72^\circ$ (CORRNIF $R^2 = 0.9907$), and an illus-	216
trative loss-aware Maxwell-Garnett birefringence screening (Section 4, Q3 and Q4).	217
	218
	219
	220
	221
	222
	223
	224
	225
	226
	227
	228
	229
	230

Section 5 discusses implications for data-driven GLAD modelling and the limits of effective-medium optical screening.

2 Computational Methods

2.1 Simulation framework

Figure 1 summarises the mechanism before the formal description below: an experimentalist setting the deposition angle α on a GLAD chamber is directly setting the tilt of the incoming atom beam relative to the substrate normal, and the programmed substrate rotation is what turns this into a helix rather than a simple tilted column.

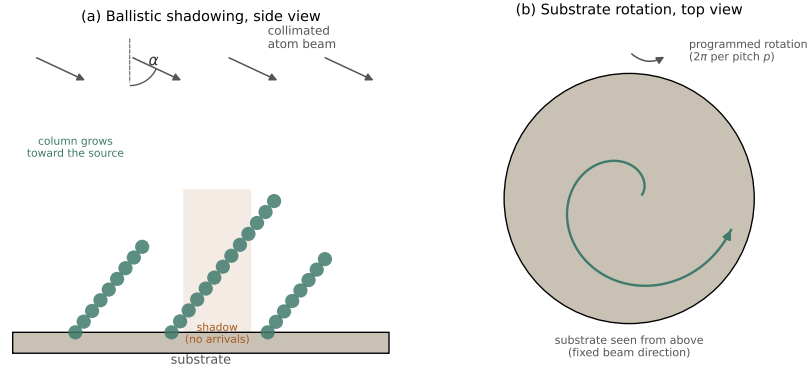


Fig. 1 The two knobs an experimentalist turns on a GLAD chamber – beam tilt α and substrate rotation – are exactly the two inputs to this simulator; everything else (columns, shadowing, voids) is an emergent consequence of these two settings plus first-contact sticking. Schematic of the ballistic deposition mechanism (conceptual diagram, not simulation output; compare Figure 2 for the simulator’s actual output). (a) Side view: atoms arrive as a collimated beam at angle α from the substrate normal and stick irreversibly on first contact, with no subsequent surface diffusion. A column that nucleates first casts a geometric shadow (shaded region) that suppresses further arrivals immediately behind it – this is the entire origin of the void fraction trend in Section 3.1. (b) Top view: the substrate rotates continuously during growth, tracing one full 2π turn per helical pitch p , which is what makes the resulting columns helical rather than straight.

Physically, the simulator asks a simple question: if atoms travel in straight lines from the source and stop irreversibly the instant they touch another atom or the substrate—i.e. if the only thing shaping the film is geometric shadowing, with no atom motion after landing—how much of the deposited volume ends up empty (void)

rather than filled (solid), and how does that depend on the deposition angle α that an experimentalist sets on the chamber? Simulations were performed with a custom bead-sphere ballistic deposition code. Atoms are launched from a collimated beam incident at angle α from the substrate normal. Each atom follows a straight-line (ballistic) trajectory and sticks irreversibly to the first atom it contacts; no surface diffusion, relaxation, or evaporation is included. The film grows on a square substrate of lateral size $L_x \times L_y = 100 \text{ nm} \times 100 \text{ nm}$ with periodic (minimum-image) boundary conditions in the two in-plane directions. The model contains no material-specific physics: “Cu” enters only through the hard-sphere atomic radius (i.e. a Cu-scale length), not through any Cu-specific diffusion, sticking, or energetic parameter. The simulator is therefore a *generic hard-sphere ballistic helical GLAD model parameterised at a Cu atomic scale*, and the geometric outputs apply to any material only insofar as its real morphology is shadowing-dominated; the later substitution of tungsten optical constants (Section 4.5) is an illustrative material substitution, not a calibrated W-film model.

2.2 Helical substrate and parameter space

Physically, this subsection reproduces in simulation the two knobs an experimentalist actually turns during a helical GLAD run: the fixed deposition angle α (substrate tilt relative to the vapour beam) and the programmed substrate rotation that traces out one helical turn per pitch length p ; the goal is to isolate how much of the resulting morphology change is due to α alone, holding the rotation program and target film height fixed. A helical hard-sphere nanostructure template is defined with target height $h = 300 \text{ nm}$ and helical pitch $p = 150 \text{ nm}$ (two full turns within the film). The pitch is a controlled numerical reference derived from the standard helix pitch relation using the vertical-growth-rate convention; it is used here to isolate deposition-angle dependence

and is not presented as an experimentally calibrated Cu pitch. Twelve configurations are simulated for the main morphology and reproducibility study: deposition angles $\alpha \in \{60, 65, 70, 75, 80, 80, 80, 85, 85, 85, 87, 89\}^\circ$, where three configurations at $\alpha = 80^\circ$ and three at $\alpha = 85^\circ$ use independently seeded random-number generators to assess stochastic reproducibility. A thirteenth simulation at $\alpha = 72^\circ$ was subsequently performed exclusively as a prospective validation case for the CORRNIIF interpolation (Section 4.3); it was excluded from CorrNIF training and from all main-dataset descriptive analyses. The deposition angle α is measured from the substrate normal, and the helix is generated by programmed azimuthal rotation of the substrate during growth (one 2π azimuthal turn per pitch length p). Table 1 collects the simulation and analysis parameters for reproducibility; quantities not separately parameterised in this ballistic model (beam angular divergence, arrival-position distribution beyond uniform random azimuth, explicit nucleation/sticking rules) are defined by the simulation code (available from the corresponding author) rather than by tunable inputs.

Before the quantitative characterisation that follows, Figure 2 shows directly what the simulator produces: real voxelised cross-sections of a simulated film, not a schematic. At $\alpha = 89^\circ$, the lateral (xy) slice shows isolated, well-separated columns rather than a continuous film, and the vertical (xz) slice shows these columns as tilted streaks – the geometric shadowing signature that the tangent rule and the void-fraction trend in Section 3.1 both quantify. At $\alpha = 60^\circ$, the same lateral slice is nearly fully covered, consistent with the much lower void fraction at this shallower angle.

2.3 Void-fraction definition

Physically, this quantity is the porosity-like measure of the film: the fraction of the deposited volume that is empty space between column atoms rather than solid material, computed geometrically from the known hard-sphere size and count of the deposited atoms rather than from an imaging or scattering measurement. The

Table 1 Simulation and analysis parameters. Quantities marked “code” are fixed by the ballistic simulator code rather than exposed as tunable inputs.

Parameter	Value	Source/status
<i>Geometry and domain</i>		
Substrate lateral size $L_x \times L_y$	100 nm $\times$ 100 nm	prescribed in this work
Lateral boundary conditions	periodic (minimum-image), XY	simulator setting
Target film height h	300 nm	prescribed in this work
Final height range (12 runs)	300.0 nm–300.1 nm	simulation result, this work
Helical pitch p (turns)	150 nm (2 turns in h)	controlled numerical reference; not Cu-calibrated
Deposition-angle convention	polar, from substrate normal	simulator convention
Azimuthal rotation law	programmed, 2π per pitch length	derived from prescribed pitch
<i>Ballistic deposition model</i>		
Hard-sphere radius R (diameter $2R$)	0.1 nm (0.2 nm)	Cu-scale model parameter (optical screening subsection uses the recalibrated $R = 0.128$ nm campaign; see Section 4.5)
First-contact criterion	$2R$ physical contact	simulator rule
Sticking probability	1 (irreversible, first contact)	simulator rule
Surface diffusion / re-emission / evap.	none / none / none	prescribed model scope
Re-emission, angular divergence	none; collimated beam	prescribed model scope
XY launch distribution	uniform random over the cell	simulator rule
Initial seed layer	10^4 seed atoms	numerical setting, this work
Nominal deposition rate	0.2 nm/s (dense-equiv.)	numerical setting, this work
Substrate temperature	not active in the ballistic physics	not used by this model
“Cu” material content	hard-sphere radius (a Cu-scale length) only	not a calibrated Cu model
Deposited atoms per run	1.1×10^7 – 1.6×10^8	simulation result, this work
Stopping criterion	target height $h = 300$ nm reached	prescribed in this work
Random seeds (RNG)	1 per angle; 3 at $80^\circ, 85^\circ$	numerical setting, this work
<i>Voxelisation and correlation</i>		
Voxel grid	$50 \times 50 \times 151$ (2 nm XY, 2 nm Z)	numerical setting, this work
Voxelisation kernel	Gaussian bead-centre histogram	analysis setting
Kernel / density normalisation	max-normalised, $\rho_{\text{voxel}} \in [0, 1]$	analysis setting
Occupancy threshold	$\rho_{\text{voxel}} > 0.5$	analysis setting
Derotation interpolation	bilinear, periodic wrap	corrected analysis setting
Correlation radial binning	integer-voxel shells	analysis setting
Correlation radial range	$r \leq 24 \text{ vox} \approx L/2$ (trusted $\lesssim 10 \text{ vox} \approx 20 \text{ nm}$)	finite-domain analysis choice
<i>Illustrative optical screening</i>		
Optical wavelength λ	≈ 500 nm	prescribed screening wavelength
Column dielectric (illustrative, W)	$\epsilon_W = 4.41 + 19.60i$	Ref. [28]
Host dielectric	$\epsilon_h = 1$ (vacuum)	prescribed screening host
Optical film thickness	300 nm	same prescribed film height

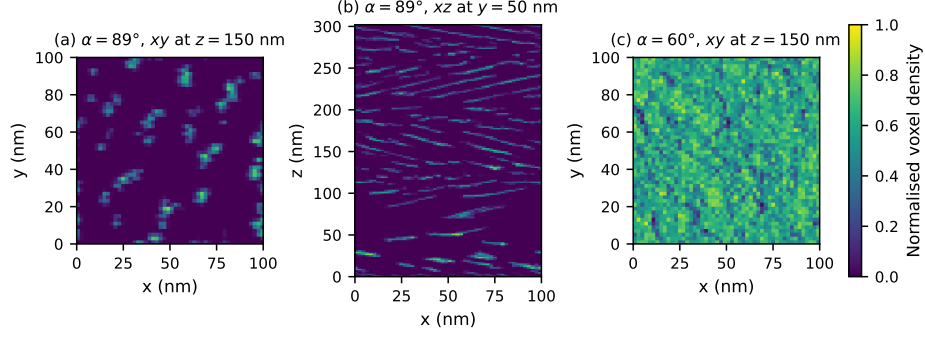


Fig. 2 Direct visual evidence of the shadowing mechanism that the rest of this paper characterises quantitatively: isolated, tilted columns at high α versus a near-continuous film at low α . Voxelised density cross-sections of simulated Cu GLAD films ($h = 300$ nm, pitch = 150 nm, seed 1, 2 nm voxels). (a) Lateral (xy) slice at mid-height ($z = 150$ nm), $\alpha = 89^\circ$: isolated columns, 8.2% occupied. (b) Vertical (xz) slice at $y = 50$ nm, $\alpha = 89^\circ$: the same columns appear as tilted streaks, the direct signature of ballistic shadowing along the deposition direction. (c) Lateral slice at $\alpha = 60^\circ$ for comparison: 83.5% occupied, a near-continuous film at this shallower angle. Despite occupying only 8.2% of the volume at $\alpha = 89^\circ$, the column network remains fully percolated (largest connected fraction = 0.974, i.e. essentially all deposited material belongs to one connected structure spanning the film).

bead-sphere void fraction is defined as:

$$P = 1 - \frac{NV_b}{L_x L_y z_{\max}}, \quad V_b = \frac{4}{3}\pi R^3, \quad (1)$$

where N is the number of deposited atoms, R is the atom radius, and $z_{\max}$ is the height of the topmost atom. This quantity directly corresponds to the open volume in the idealised hard-sphere packing and is logged by the simulator at the end of each simulation run. It differs from the voxel-mean quantity $1 - \langle \rho_{\text{voxel}} \rangle$ used for density-field analyses (Sections 3.2 and 3.3).

2.4 Voxel density field

Physically, this step turns the raw list of atom positions into something directly comparable to a depth-resolved microstructure map, analogous to a reconstructed FIB-SEM or nano-CT tomogram of a real film, so that how dense or open the structure is at

a given depth and lateral position can be tracked separately from the single, film-averaged void fraction of Section 2.3. For depth-resolved and PCA analyses, atom positions are projected onto a $50 \times 50 \times 151$ voxel grid (2 nm XY resolution, 2 nm Z resolution, spanning the $100 \text{ nm} \times 100 \text{ nm} \times 300 \text{ nm}$ cell) using a Gaussian bead-centre kernel. The resulting normalised density field $\rho_{\text{voxel}}(\mathbf{r}) \in [0, 1]$ encodes the local atom-sphere overlap fraction, with the continuous three-dimensional material distribution sampled on a regular spatial grid as in the tomographic analogy above. The Level-A run at $\alpha = 60^\circ$ had a corrupted final checkpoint (active-slab resume duplication artefact, physically impossible atom count of 163 M vs. the expected ~ 85 M at 300 nm); both its bead-sphere value ($P = 77.2\%$) and its voxel are invalid and excluded from all analyses. The openness-trend curve uses $P = 88.1\%$ from a clean single-shot replacement run (literature-matched benchmark, independent simulation from a validated starting state; SHA-256 c7509b21...).

2.5 Morphological descriptors and PCA

Physically, a single voxel field contains far more information than an experimentalist can compare across many deposition angles at a glance; the goal here is to compress each film down to a handful of numbers that capture how open it is, how directionally lopsided its columns are, and whether the columns broaden or narrow with depth, and then to check whether deposition angle alone organises these numbers into a simple, low-dimensional trend. For each of the eleven valid voxel files we use three non-redundant descriptors: (i) openness proxy $1 - \langle \rho_{\text{voxel}} \rangle$; (ii) XY anisotropy ratio (ratio of power along orthogonal in-plane axes); (iii) apparent width-gradient proxy (linear fit to column-width growth with depth; used only as a descriptive feature, not interpreted as a physical fanning exponent). We deliberately exclude the mean filling fraction $\langle \rho_{\text{voxel}} \rangle$, which is exactly complementary to the openness proxy (openness = $1 - \langle \rho_{\text{voxel}} \rangle$): including both perfectly collinear measures double-counts the

density axis and artificially inflates the variance attributed to it on the leading component. Features are standardised before PCA; principal components are computed via SVD, and variance fractions and loadings are reported. The eleven observations include repeated seeds at 80° and 85° , which we keep as individual points (rather than averaging) so that the seed-to-seed scatter is visible. The result is therefore an exploratory descriptor-space summary, not a demonstration of a general morphology manifold.

2.6 Statistical morphology pipeline

Corotating-frame pair correlation

Physically, this descriptor asks how likely a column is to be found at a given lateral distance from another column, once the steady rotation of the substrate has been numerically “undone” so that a static column arrangement, rather than a rotating helix, is being compared from film to film; it is the real-space analogue of the column-spacing information that a scattering experiment (SAXS/GISAXS) would extract from the diffraction pattern of the same film. The corotating-frame correlation starts from the binary occupancy field $I(\mathbf{r}) = [\rho_{\text{voxel}} > 0.5]$, from which the filling fraction $f = \langle I \rangle$ is taken. Each z -slice $I(\cdot, \cdot, k)$ is then derotated by the local helix phase $\phi(k) = 2\pi k/p_{\text{vox}}$, where $p_{\text{vox}} = p/\Delta z$ is the pitch in voxels, using bilinear interpolation to avoid the pixel-level jitter that nearest-neighbour resampling accumulates over the 151 z -slices. Because the domain is periodic, the bilinear resampling uses periodic (wrap) source coordinates rather than edge-clamped coordinates; the clamped variant violates per-slice mass conservation by up to 10% near the boundary and is therefore replaced here. Switching from the edge-clamped to the periodic-wrap derotation perturbs h_g by a mean absolute difference of 0.007 over the full curve (0.015 at short range, where the steep near-origin drop amplifies small horizontal shifts; maximum pointwise 0.035). This is comparable to the intrinsic seed-to-seed scatter (≈ 0.008 ,

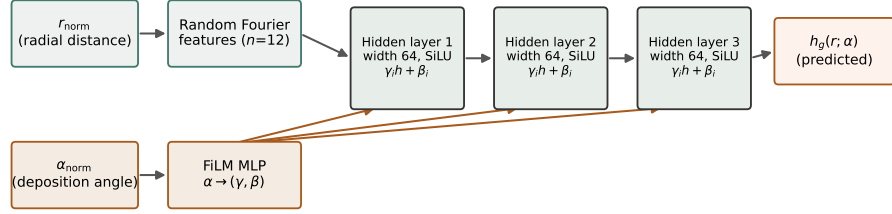
below) and leaves the angle ordering and every qualitative conclusion unchanged (validation tests included with the analysis code). Bilinear derotation maps the binary slice to a *continuous* occupancy field $J \in [0, 1]$, for which the binary identity $\langle J^2 \rangle = \langle J \rangle$ no longer holds; we therefore normalise by the empirical zero-lag (continuous-covariance) variance rather than by the binary value $1/f$ (Eqs. 3–4). The circular autocorrelation $C(r) = \langle J(\mathbf{x})J(\mathbf{x}+r) \rangle$ is obtained by FFT, radially averaged, and the result is averaged over all z -slices to yield the ensemble-mean corotating-frame curve $\langle h_g(r; \alpha) \rangle_z$. This is computed for the seven single-seed configurations $\alpha \in \{65, 70, 75, 80, 85, 87, 89\}^\circ$. The lateral domain is 100 nm with periodic boundaries, and the radial average is taken to $r_{\max} \approx 24$ voxels (≈ 48 nm), i.e. close to half the box width $L/2$. Under the periodic (minimum-image) convention the correlation near $r \rightarrow L/2$ is constrained by the finite domain, so the curves are informative mainly at short range ($r \lesssim 10$ voxels ≈ 20 nm, the regime that dominates the descriptor and the emulator targets) and the long-range tail should be read with caution. A box-size convergence study (e.g. 100, 150, 200 nm at representative angles) would be required to confirm that the short-range correlation is free of finite-size or periodic artefacts; this requires new simulations and is left as future work. (A related but distinct check – bulk atom-density convergence with box size at fixed pitch, not the correlation function itself – was performed and found NOT converged at $L = 100$ nm; see the box-size limitation in Section 5.4.)

CorrNIF architecture and training

CORRNIF is a compact parametric emulator that maps deposition angle α and radial distance r to the predicted pair-correlation value $h_g(r; \alpha)$ at any combination of these two inputs—analogous to a parametric dispersion model that evaluates a refractive-index spectrum at any wavelength and filling fraction without re-running a full electromagnetic simulation.

A compact neural implicit field (CORRNIF) maps $(\alpha_{\text{norm}}, r_{\text{norm}}) \rightarrow h_g(r; \alpha)$, where $\alpha_{\text{norm}} = (\alpha - 74.5^\circ)/14.5^\circ$ and $r_{\text{norm}} = r/r_{\max}$. The radial coordinate is

599 encoded via $n_{\text{freq}} = 12$ random Fourier features [26]: $\mathbf{h}_0 = [\sin(2\pi\mathbf{B}r), \cos(2\pi\mathbf{B}r)]$,
600 $\mathbf{B} \sim \mathcal{N}(0, \sigma_r^2)$, $\sigma_r = 2$. Three fully connected hidden layers (width 64, SiLU acti-
601 vation) receive angle conditioning (feature-wise linear modulation, FiLM) from α :
602 $\mathbf{h}_{i+1} = \text{SiLU}[(1+0.1\gamma_i)\mathbf{W}_i\mathbf{h}_i+0.1\beta_i]$, where (γ_i, β_i) are produced by an α -conditioned
603 two-layer MLP (width 32, ReLU). Figure 3 summarises this architecture.



FiLM: each hidden layer's activation is rescaled by $\gamma_i(\alpha)$ and shifted by $\beta_i(\alpha)$ before the nonlinearity — the network's function of r changes continuously with α , the same way a parametric dispersion model is evaluated at any wavelength without refitting.

619 **Fig. 3** The angle conditioning does not add a third input alongside r ; it reshapes the network's
620 response to r at every layer, which is why a single trained model can be evaluated smoothly at
621 angles it never saw during training (Section 4.3). Schematic of the CORRIF architecture (conceptual
622 diagram). The radial coordinate r is Fourier-encoded and passed through three FiLM-conditioned
623 hidden layers; the deposition angle α does not enter as a third raw input but instead rescales (γ_i) and
624 shifts (β_i) each layer's activation via a small side network, analogous to how a parametric dispersion
625 model is evaluated at any wavelength without being refitted at that wavelength.

627 Training uses the Adam optimiser (learning rate 10^{-3} , 1000 epochs, batch size 512,
628 MSE loss on h_g). For the corrected analysis reported here, CORRIF was retrained
629 from scratch on the periodic-wrap, continuous-covariance target; ten independent neu-
630 ral initialisations were run for each validation fold and for the prospective 72° case.
631 Generalisation is evaluated by a seven-fold leave-one-angle-out (LOOA) protocol: in
632 each fold, CORRIF is trained on six deposition angles and used to predict $h_g(r; \alpha_{\text{held}})$
633 without any adaptation on the held-out angle—analogous to fitting an optical disper-
634 sion model to spectra measured at six incidence angles and evaluating its prediction
635 at a seventh, held-out angle.

Maxwell-Garnett effective-medium model

Physically, this step asks what optical anisotropy a film with the simulated column filling fraction $f(\alpha)$ would show *if* its electromagnetic response could be approximated by smoothly averaging the column and void dielectric properties together, rather than by resolving the actual column geometry wave-optically—a standard first estimate of form birefringence, used here as a scoping tool rather than a precision prediction. Effective optical properties are derived from the Maxwell-Garnett (MG) effective-medium approximation for *aligned* cylindrical inclusions in a host medium [27] (Figure 4), a special case of the broader family of mixing formulas surveyed by Sihvola [29] and related to the symmetric effective-medium approximation of Bruggeman [30];

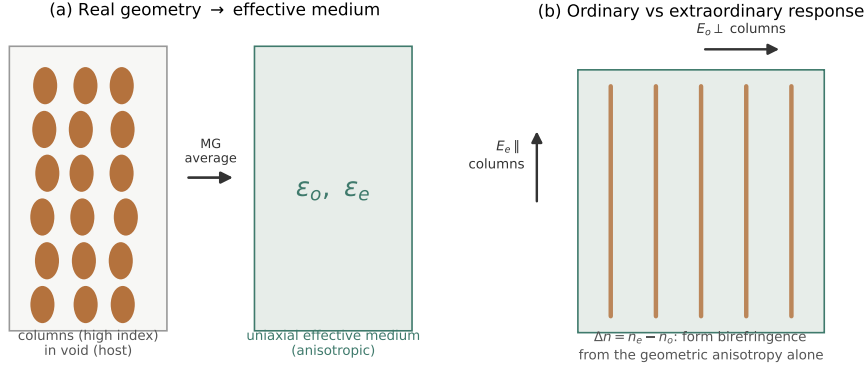


Fig. 4 The birefringence reported below is not an extra material property being added to the model; it falls directly out of averaging an already-anisotropic column geometry into a single effective medium. Conceptual illustration of the Maxwell-Garnett effective-medium approximation (not simulation output). (a) Aligned high-index cylindrical inclusions (columns) embedded in a low-index host (void) are replaced by a single uniaxial effective medium with two principal dielectric constants, ϵ_o and ϵ_e . (b) The two principal optical axes: the extraordinary axis E_e lies parallel to the columns, the ordinary axis E_o perpendicular to them. Form birefringence $\Delta n = n_e - n_o$ arises purely from this geometric anisotropy, before any intrinsic material birefringence is considered.

empirical extensions of this approach specifically fitted to the measured principal refractive indices of obliquely deposited oxide sculptured thin films (Ta_2O_5 , TiO_2 ,

691 ZrO_2) are given by Hodgkinson et al. [31], building on the birefringent-thin-film for-
 692 malism of Hodgkinson and Wu [32]. This describes a uniaxial medium with the optic
 693 axis along the column direction; it is a local, aligned-column approximation and
 694 does *not* capture the rotating optic axis, circular birefringence, or circular dichroism
 695 of a true helical (chiral) film, nor pitch-scale resonances or spatial dispersion. The
 696 quasi-static homogenisation underlying MG requires the structural period to be small
 697 compared with the wavelength; here the helical pitch $p = 150 \text{ nm}$ and $\lambda \approx 500 \text{ nm}$
 698 give $p/\lambda \approx 0.30$, which is not deep in the quasi-static limit, so the calculation should
 699 be read as an order-of-magnitude screening estimate of the *local* form birefringence of
 700 an aligned columnar array rather than as the full optical response of the helical archi-
 701 tecture. As a sensor-design hypothesis-generation illustration only, column dielectric
 702 functions are taken from tungsten ($\epsilon_W = 4.41 + 19.60i$ at $\lambda \approx 500 \text{ nm}$, corresponding
 703 to optical constants $n = 3.5$, $k = 2.8$ [28]) in a vacuum host ($\epsilon_h = 1$). Both the ordi-
 704 nary (ϵ_o , transverse polarisation) and extraordinary (ϵ_e , longitudinal, Wiener parallel
 705 limit) dielectric functions (Eqs. 5 and 6) are evaluated as complex quantities, and the
 706 effective indices are obtained as $N = \sqrt{\epsilon}$ on the passive (positive-imaginary) branch,
 707 so that both refractive index n and extinction k are retained for each polarisation.
 708 The filling fraction $f(\alpha)$ is *not* a unique material-independent quantity: the ballistic
 709 geometry yields two distinct measures (the bead-sphere solid fraction $1 - P$ and the
 710 thresholded voxel occupancy $f_{\rho>0.5}$) that differ by up to a factor of four (Section 4.5).
 711 Real GLAD morphology is moreover material-dependent through surface diffusion,
 712 sticking, re-emission, homologous temperature, deposition rate, and pressure; the sub-
 713 stitution of W (or Mo, Co) optical constants into the Cu-scale geometry is therefore
 714 an illustrative material substitution, not a prediction of a realisable W film, and any
 715 quantitative transfer would require material-specific calibration.
 716
 717
 718
 719
 720
 721
 722
 723
 724
 725
 726
 727
 728
 729
 730
 731
 732
 733
 734
 735
 736

3 Results

3.1 Bead-sphere void fraction and reproducibility

This subsection addresses the reproducibility half of **Q1** (the comparison against published optical void fractions, the other half of Q1, follows in Section 3.4). Figure 5 shows P as a function of deposition angle for all twelve main configurations. The void fraction increases monotonically from $P = 88.1\%$ at $\alpha = 60^\circ$ (clean benchmark run; Level-A primary excluded, see Section 2.4) to $P = 98.5\%$ at $\alpha = 89^\circ$, consistent with the expected shadowing-enhancement at high obliquity.

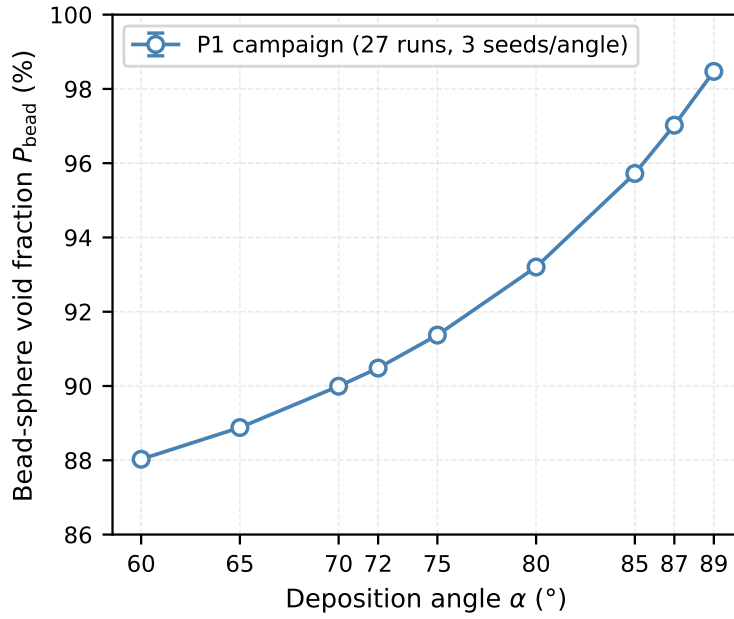


Fig. 5 Steeper deposition angles produce systematically more open (porous) films, and this trend is highly reproducible across independent random seeds. Bead-sphere void fraction P (Eq. 1), mean $\pm$ standard deviation across three independent random seeds, as a function of deposition angle for the authoritative P1 campaign (27 runs: nine angles $\alpha \in \{60, 65, 70, 72, 75, 80, 85, 87, 89\}^\circ$, three seeds each; $h = 300$ nm, pitch = 150 nm). The void fraction increases monotonically with α , reaching 98.5% at $\alpha = 89^\circ$; run-to-run reproducibility at fixed α is discussed quantitatively in Table 2.

Three-seed reproducibility tests at $\alpha = 80^\circ$ and $\alpha = 85^\circ$ yield coefficients of variation in atom areal density of $\text{CV}_{\rho_A} = 0.19\%$ and 0.23% , respectively (Table 2).

783 The bead-sphere void fractions at both angles are identical to one decimal place across
784 all three seeds, demonstrating that the ballistic simulator exhibits excellent stochastic
785 reproducibility.
786

788 **Table 2** Three-seed reproducibility at $\alpha = 80^\circ$ and $\alpha = 85^\circ$. $\bar{P}$:
789 mean bead-sphere void fraction; σ_P : standard deviation; CV_{ρ_A} :
790 coefficient of variation of atom areal density.

α ($^\circ$)	n seeds	$\bar{P}$ (%)	σ_P (pp)	$\bar{\rho}_A$ (nm^{-2})	CV_{ρ_A} (%)
80	3	93.2	< 0.1	4866	0.19
85	3	95.7	< 0.1	3063	0.23

795 $\bar{\rho}_A$ is the deposited-atom areal density (total atoms per
796 unit substrate area, integrated over the film height):
797 per-seed values 4870.9/4854.8/4871.3 nm^{-2} at 80° and
798 3057.2/3060.1/3070.7 nm^{-2} at 85° . All three seeds give P iden-
799 tical to 0.1 pp.
800

805 3.2 Depth-resolved void-fraction profiles

806
807 This subsection addresses the depth-variation part of **Q2**. Figure 6 shows the layer-
808 wise void fraction $1 - \bar{\rho}_z(z)$ (mean over the 50×50 XY grid at each Z-slice) for all
809 eleven valid configurations.
810

811 Three features are evident:

- 812 1. *Dense substrate layer*. All configurations show a lower void fraction near the
813 substrate ($z < 30 \text{ nm}$), reflecting the initial random-accumulation regime before
814 columnar shadowing develops. This layer is thinner and less dense at higher α .
815
- 816 2. *Monotonic mid-film void fraction*. The bulk columnar region (approximately $z =$
817 $50 \text{ nm} - h_{\text{top}}$) shows a void fraction that increases with α , consistent with stronger
818 shadowing. At $\alpha = 65^\circ$ the mid-film layer void fraction is approximately 57%,
819 whereas at $\alpha = 89^\circ$ it approaches 95%.
820
821
822
823
824
825
826
827
828

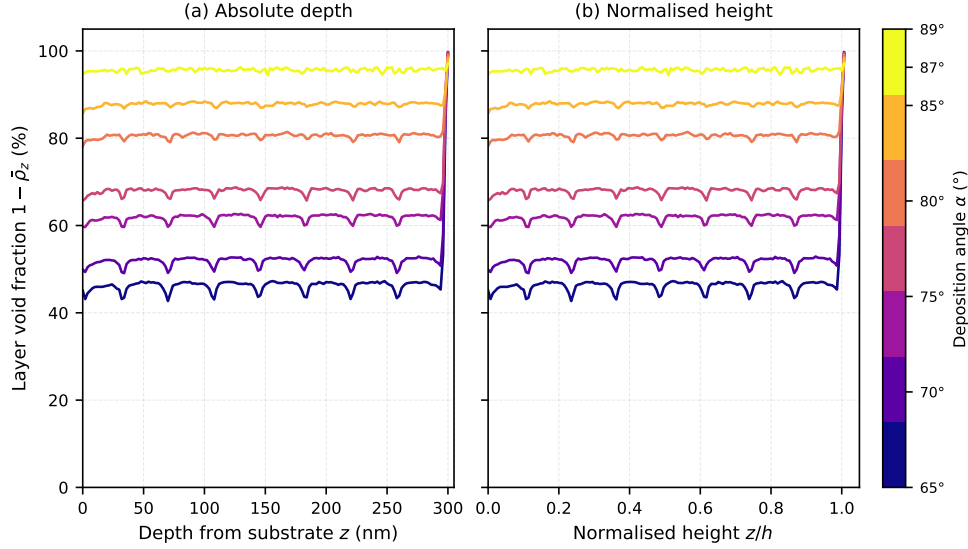


Fig. 6 Every film has a denser layer near the substrate before opening up into the angle-dependent shadowing regime that dominates the bulk of the growth. Depth-resolved void fraction $1 - \bar{\rho}_z$ as a function of (a) absolute depth from substrate (z) and (b) normalised film height (z/h), for eleven helical Cu GLAD configurations spanning $\alpha = 65^\circ$ – 89° (seed 0 for each angle; $\alpha = 60^\circ$ excluded due to a voxelisation artefact at this angle). Curves are colour-coded by α (plasma colourmap, 65° = violet, 89° = yellow). The quantity $1 - \bar{\rho}_z$ is the smeared-kernel void fraction averaged over the 100×100 nm² substrate plane at each depth slice, which differs from the global bead-sphere P . All curves show a dense substrate layer ($z < 30$ nm) transitioning to a shadowing-dominated bulk regime.

3. *Normalised-height convergence.* When height is normalised by film top z/h (panel b), the mid-film profiles collapse more closely. Because all films share essentially the same target height ($h \approx 300$ nm), this normalisation cannot separate a growth-kinetics origin from an absolute-height origin; a height series at fixed α (available in the levelB dataset but not analysed here) would be required to attribute the collapse to growth kinetics, and we report it only as an observed collapse. The profiles also show low-amplitude oscillations with depth; these are consistent with the helical rotation period and the finite voxel slice spacing and are not interpreted as a distinct physical growth modulation pending a dedicated sampling-convergence check.

These depth gradients are relevant to optical and sensor properties, which depend on the local void-fraction profile rather than the bulk average, although a quantitative optical use would require the convergence checks noted above.

3.3 Morphology latent space

This subsection addresses the low-dimensional-summary part of **Q2**. Figure 7 shows the PCA of the three non-redundant morphological descriptors extracted from the eleven valid voxel fields.

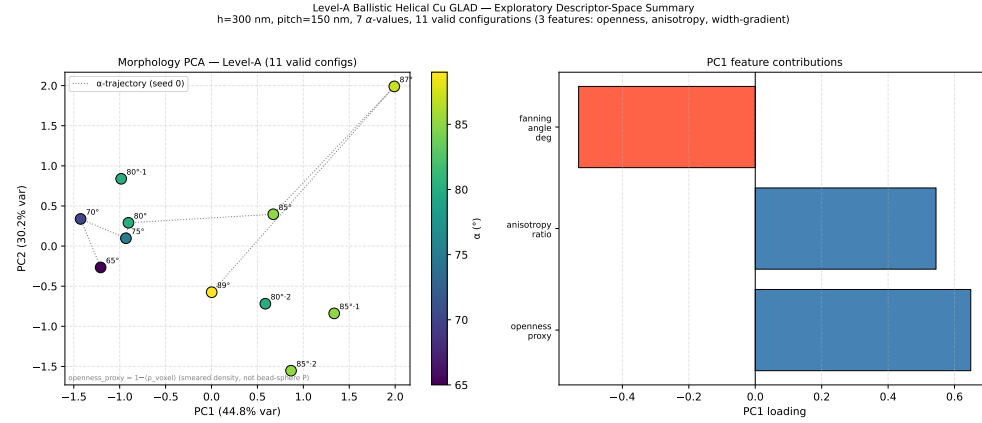


Fig. 7 Deposition angle is the dominant organising variable in this three-descriptor morphology space, but repeated seeds at the same angle still scatter visibly, showing that only the bulk density trend, not the full morphology, is tightly reproducible. (a) Principal component analysis (PCA) of three non-redundant morphological descriptors (openness $1 - \langle \rho \rangle$, anisotropy ratio, apparent width-gradient proxy) for the eleven valid ballistic helical Cu GLAD voxel fields ($\alpha = 60^\circ$ excluded due to a voxelisation artefact; the mean filling fraction is omitted as it is exactly complementary to openness). Points are colour-coded by deposition angle α ; the dotted line connects the seed-0 points in order of increasing α , and the three seeds at $80^\circ/85^\circ$ appear as separate points to show the seed-to-seed scatter. The first two principal components capture $44.8 + 30.2 = 75\%$ of the descriptor variance in this exploratory three-feature summary (with three features the per-component null expectation is 33%, so PC1 is only moderately dominant). (b) PC1 feature loadings: all three descriptors contribute comparably (openness 0.65, anisotropy 0.54, width-gradient -0.53), so PC1 is a blend rather than a single-descriptor axis.

With the redundant filling fraction removed, the first two principal components account for $44.8 + 30.2 = 75.0\%$ of the descriptor variance. PC1 is now a balanced blend of all three descriptors (loadings: openness 0.65, anisotropy 0.54, width-gradient -0.53)

rather than being dominated by the density measures, and PC2 is carried mainly by anisotropy and the width-gradient (-0.69 and -0.72 , openness ≈ 0). The points show a broad shift toward larger PC1 as α increases, but the trajectory is not a clean monotonic locus: the three seeds at 80° and 85° are visibly scattered in the PC1–PC2 plane. This is consistent with the reproducibility analysis of Section 3.1, where the bulk density is reproducible to sub-percent level but the higher-order descriptors (anisotropy, width gradient) carry more run-to-run variability. Within this fixed-height, fixed-pitch sweep, deposition angle is the dominant *controlled* source of descriptor variation, but—given only eleven observations and three features, several of them repeated seeds—we read this as an exploratory summary rather than evidence of a smooth low-dimensional morphology manifold.

3.4 Comparison with literature void fractions

This subsection addresses the second half of **Q1**: the quantitative comparison against published optical void fractions. Figure 8 compares the ballistic bead-sphere void-fraction curve with published optical void fractions for GLAD metal films at $\alpha = 85^\circ$ from Fischer & Schubert (2013).

At $\alpha = 85^\circ$, the ballistic simulation yields $P = 95.7\%$, while the optical void fractions of GLAD metal films (Co: $75.38 \pm 0.01\%$; Ti: $78.30 \pm 0.04\%$; NiFeMo: $75.0 \pm 0.1\%$) cluster around a median of $\approx 75\%$, giving a systematic offset of $\Delta P = +20.7$ pp.

This offset is expected for three reasons:

1. *Absence of surface diffusion.* Ballistic models produce more open structures than real deposited films, where adatom mobility drives local densification after impact [33]; oblique-angle-deposited films with active adatom mobility show measurably lower porosity and altered deposition-rate scaling than a purely geometric shadowing prediction would give [34], and this densification is known to broaden individual GLAD Cu nanorods anisotropically relative to the ballistic-limit column shape [35].

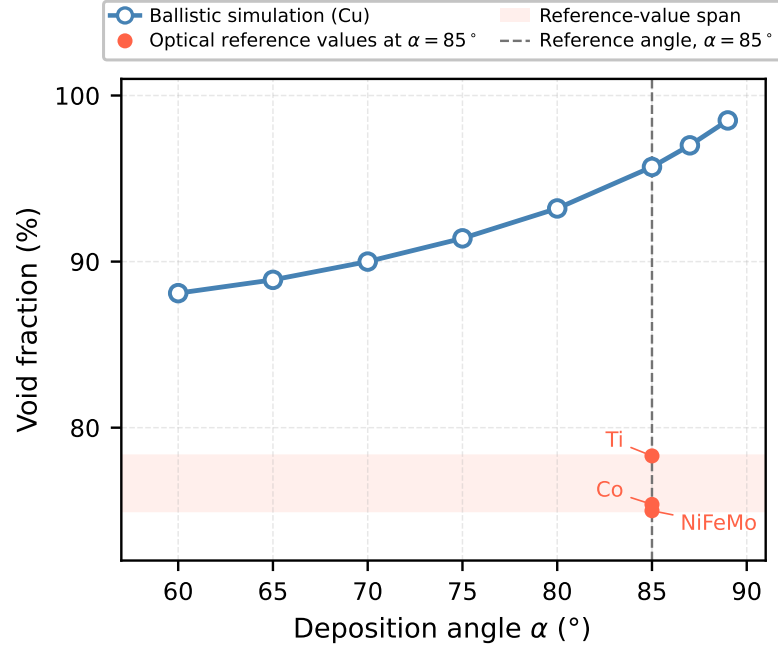


Fig. 8 The purely geometric ballistic model is systematically more open than real deposited GLAD metal films, confirming that surface diffusion and measurement-convention differences (not captured by this model) matter quantitatively. Comparison of bead-sphere void fraction P (ballistic Cu GLAD simulation, blue curve) with published optical void fractions for GLAD metal films (Co: 75.4%; Ti: 78.3%; NiFeMo: 75.0%) at $\alpha = 85^\circ$ measured by the anisotropic Bruggeman effective-medium approximation (ABEMA method; Fischer & Schubert 2013 [10]). Red points with error bars: individual published values; shaded band: literature range. The ballistic simulation overestimates the published median by +20.7 percentage points at $\alpha = 85^\circ$. The three contributing sources of mismatch (absence of surface diffusion, measurement-convention difference, and material difference) are discussed in Section 5.4.

2. *Measurement-convention mismatch.* The bead-sphere formula counts volume occupied by rigid spheres, while optical ABEMA infers void fraction from effective-medium theory applied to ellipsometric data; the two conventions do not coincide exactly even for the same physical pore structure.
3. *Material mismatch.* The literature values are for Co, Ti, and NiFeMo; Cu has a different diffusion length and sticking coefficient, so a direct quantitative comparison would require Cu-specific experiments (structure-conduction property relations for a related Cu-based nanostructured system are characterised in El Beainou [36]).

Taken together, these three mismatches preclude a direct quantitative comparison. The +20.7 pp difference observed at $\alpha = 85^\circ$ should be read as an *indicative upper-bound offset* rather than a calibrated correction: Cu-specific GLAD void-fraction measurements would be required to establish a material- and method-consistent calibration reference.

A Cu-specific angle-dependent porosity series is in fact available for a related material system: Ben Nacer et al. [37] report an optically-inferred (Bruggeman effective-medium) porosity for GLAD Cu_2O thin films of 23%, 28%, 33%, and 37% at $\alpha = 0^\circ, 60^\circ, 75^\circ$, and 85° respectively, likewise increasing monotonically with α . These numbers are an order of magnitude smaller than, and not on the same scale as, the bead-sphere void fraction reported here (88.1–98.5% over $\alpha = 60\text{--}89^\circ$): the two are different physical quantities by construction, not two measurements of the same one. The present simulation reports an idealised geometric packing fraction of non-diffusing, non-reacting hard spheres with no material chemistry, whereas the Ben Nacer et al. value is an effective-medium fit to ellipsometric data from a real, post-deposition-annealed, oxidised (Cu_2O) film subject to surface diffusion, oxidation, and densification processes entirely absent from the model. No quantitative agreement in magnitude is expected or claimed between them, for the same reasons detailed for the Fischer & Schubert comparison above. The only comparison that is scientifically meaningful between these two datasets is that both quantities increase monotonically with deposition angle over an overlapping angular range: this is a qualitative trend-level agreement, not a quantitative validation of the simulation against experiment. We therefore state only that the simulated void-fraction-versus-angle trend is qualitatively consistent with the Ben Nacer et al. Cu_2O porosity-versus-angle trend, and make no claim that the simulation validates, confirms, matches, or reproduces the experimental porosity values themselves.

1059 4 Statistical Morphology via Neural Implicit Field

1060

1061 The morphological descriptors of Section 3.3 characterise the *bulk* structure of each
1062 film. We now ask whether the *spatial* structure of the voxel density field can be mod-
1063 elled by a neural implicit field (NIF)—a compact multi-layer perceptron mapping
1064 continuous coordinates to a scalar field—and, where direct voxel prediction fails, what
1065 statistical quantity can instead be learned and interpolated across deposition angles
1066 (Q3).
1067

1071

1072

1073 4.1 Voxel-level occupancy: realization-specific maps are not

1074

1075 transferable

1076

1077 In any GLAD experiment or simulation, two runs deposited at the same angle α under
1078 identical macroscopic conditions produce the same statistical properties (void frac-
1079 tion, optical anisotropy) but different individual column arrangements in SEM images:
1080 column nucleation sites are determined by the random initial conditions of each run.
1081 The ballistic simulator exhibits the same behaviour: column positions are indepen-
1082 dently seeded by the random-number generator in every run. A model trained to
1083 predict the *absolute positions* of individual columns across independent runs is there-
1084 fore attempting to reproduce a realization-specific arrangement that is re-randomised
1085 every run—analogous to predicting the exact positions of snowflakes in two separate
1086 snowfalls at the same temperature and humidity, even though the snowflake size *dis-*
1087 *tribution* is reproducible. We test how far this limits a coordinate-conditioned model
1088 with a corotating-frame oracle experiment.
1089

1096

1097 A coordinate-conditioned neural implicit field (NIF) was trained to predict per-
1098 voxel binary occupancy $I(\mathbf{r}; \alpha) \in \{0, 1\}$ from spatial coordinates (x, y, z) and
1099 deposition angle α , using a random Fourier-feature encoder and angle condi-
1100 tioning (FiLM, feature-wise linear modulation) on α . Seven configurations ($\alpha \in$
1101

1103

1104

{65, 70, 75, 80, 85, 87, 89} $^\circ$) were trained independently, and a multi-angle leave-one-angle-out (LOOA) variant was also evaluated.

Observation: the pooled multi-angle, multi-seed production NIF does not converge below the constant-occupancy baseline. Its training binary cross-entropy stalls near $\mathcal{L}_{\text{BCE}} = \ln 2 \approx 0.693$, the value of a maximally uninformative predictor that outputs a constant probability of 0.5; for the pooled multi-angle data (overall occupancy well below one half) this is no better than—and in fact worse than—even a constant predictor set to the bulk filling fraction, i.e. the network learns nothing useful. This plateau on its own is *not* evidence that the field is unrepresentable: it can equally arise from optimisation effects (here, early stopping before convergence and contradictory occupancy labels where the three independent seeds at 80 $^\circ$ and 85 $^\circ$ are pooled), so we do not treat the production training loss as a physical result. To separate representational capacity from these confounds we ran a controlled single-realization overfit (train = test) on one deterministic field ($\alpha = 70^\circ$, balanced occupancy $f = 0.50$): the same Fourier-feature coordinate network drives the BCE from the $\ln 2$ baseline to ≈ 0 and reaches an intersection-over-union of 1.00 on a 50×50 transverse slice and 0.997 on a 24^3 sub-volume, far above the trivial all-positive baseline ($\text{IoU} = f$). The tested deterministic slice and sub-volume are therefore representable to near-perfect accuracy by the coordinate network; the production $\ln 2$ plateau is consistent with optimisation limitations (early stopping) and pooled-realization label conflicts rather than representational impossibility, although the present tests do not isolate its exact cause. The scientifically meaningful question is thus not whether one field can be fitted—the tested cases can—but whether an *absolute* occupancy map learned from one realization transfers to an independently nucleated run. To probe this we use a corotating-frame *oracle* that is not subject to optimisation failure: it measures whether the best fixed mean-template baseline—the empirical mean derotated occupancy map under the tested corotating protocol—carries any spatial information beyond the bulk

1151 filling fraction. For each α , the mean derotated occupancy map $M_{\text{train}}(x_{\text{rot}}, y_{\text{rot}})$ is
 1152 accumulated over training slices $z \in [0, 121]$ (nearest-neighbour derotation by the
 1153 local helix phase angle $\phi(k) = 2\pi k/p_{\text{vox}}$). We quantify spatial prediction success by
 1154 the column-map overlap fraction (intersection-over-union, IoU): the fraction of vox-
 1155 els correctly identified as column material by both model and reference, divided by
 1156 the union of all voxels identified as column by either. The reference baseline is the
 1157 all-positive segmentation (every voxel labelled column material), whose intersection-
 1158 over-union with the true occupancy equals the filling fraction f exactly by construction
 1159 (IoU = $|\text{truth} \cap \text{all}|/|\text{truth} \cup \text{all}| = N_{\text{true}}/N_{\text{total}} = f$). The oracle IoU gain of the
 1160 thresholded mean-template map above this baseline is:

$$\Delta = \text{IoU}_{\text{best}}(M_{\text{train}}) - f = 0.000 \quad \forall \alpha, \quad (2)$$

1167
 1168
 1169 even when evaluated *on the training region itself* (Table 3). This result holds despite
 1170 peak-to-mean ratios of M_{train} ranging from 1.25 ($\alpha = 65^\circ$) to 6.57 ($\alpha = 89^\circ$). Figure 9
 1171 shows M_{train} directly for these two extremes: at $\alpha = 65^\circ$ the map is a low-contrast, spa-
 1172 tially uncorrelated speckle field around the bulk filling fraction; at $\alpha = 89^\circ$ the higher
 1173 peak-to-mean ratio reflects only a lower, more sharply-peaked overall occupancy, not
 1174 an emergent coherent column template. Visual inspection is consistent with the zero
 1175 oracle gain: in neither case does the mean derotated map contain a spatial pattern
 1176 that a fixed-template predictor could exploit.

1177
 1178 Empirically, then, the time-averaged derotated occupancy map carries no fixed
 1179 spatial template that improves on the bulk filling fraction for *this* simulator's pooled
 1180 stochastic-nucleation protocol: the individual column positions are set independently
 1181 in each run, so a coordinate-conditioned predictor of the form $f(x, y, z, \alpha) \rightarrow I$ has
 1182 no realization-invariant target to recover beyond the uniform-occupancy baseline. We
 1183 emphasise that this is an empirical finding for the present simulator and the present
 1184
 1185
 1186
 1187
 1188
 1189
 1190
 1191
 1192
 1193
 1194
 1195
 1196

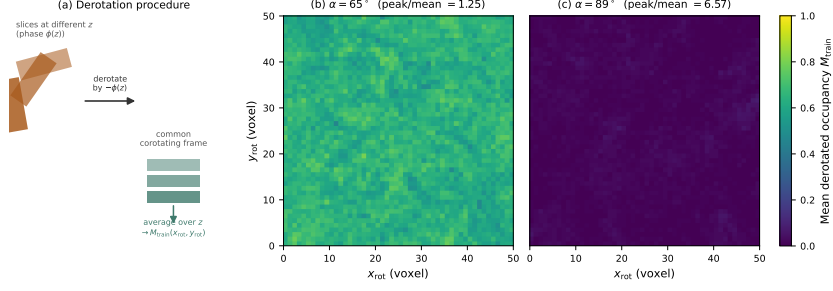


Fig. 9 The corotating-frame template itself is visually uninformative in both limits, which is why no fixed-template predictor can beat the bulk filling fraction (Table 3). (a) Schematic of the derotation procedure (conceptual diagram): each z -slice is derotated by the local helix phase $-\phi(z)$ into a common corotating frame, then averaged over z to give $M_{\text{train}}(x_{\text{rot}}, y_{\text{rot}})$. (b),(c) The resulting mean derotated occupancy map for the two extreme peak-to-mean angles, $\alpha = 65^\circ$ (peak/mean = 1.25) and $\alpha = 89^\circ$ (peak/mean = 6.57), accumulated over training slices $z \in [0, 121]$. Neither map shows a coherent, learnable spatial template: the 65° map is uncorrelated speckle around a high bulk occupancy, and the 89° map's higher peak-to-mean ratio reflects a lower overall occupancy rather than emergent structure.

corotating protocol, *not* a general proof that no model can ever fit a voxel field: a single realised field is deterministic once generated and could in principle be memorised, but such an absolute fixed-template fit would not transfer to an independently nucleated run. The result bounds only *absolute, fixed spatial templates*; it does not rule out probabilistic or generative models, latent-seed-conditioned fields, distributional prediction, or equivariant architectures that target the statistics rather than the absolute positions. The defensible conclusion is that absolute fixed spatial templates do not transfer between the tested independently nucleated realizations, whereas selected ensemble statistics (Section 4.2) are far more reproducible. In experimental terms, this is equivalent to finding that, after derotation into the corotating frame, the column positions in one GLAD film provide no template that predicts the column positions of a nominally identical film grown in a separate run.

4.2 Two-point pair correlation and CorrNIF interpolation

Since individual voxel occupancies are stochastic, we shift the target to the *ensemble-level* statistical morphology: the normalised two-point pair correlation $h_g(r; \alpha)$. This real-space descriptor is related, through Fourier transformation, to the structure-factor

Table 3 Corotating-frame oracle. M_{train} : mean derotated occupancy from training slices $z \in [0, 121]$. $\Delta_{\text{in}}/\Delta_{\text{ho}}$: IoU gain above the all-positive baseline (IoU = f) on training / holdout ($z \in [121, 151]$) regions. All gains are ≤ 0 , showing that the best fixed mean-template baseline (the mean derotated map, thresholded) does not exceed the all-positive baseline for this simulator’s pooled stochastic-nucleation protocol. $f_{\text{train}}/f_{\text{hold}}$ here use the $z \in [0, 121]/[121, 151]$ voxel-index convention of this table only; the companion study [38] reports occupied fractions over a different, full-height z -range at the same α , so its numeric values (e.g. 0.177 vs. this table’s 0.126 at $\alpha = 85^\circ$) are not directly comparable without accounting for that convention difference. An IoU gain of zero means that the best fixed spatial template does no better than the trivial all-positive segmentation—equivalent to finding that individual column positions carry no recoverable fixed spatial pattern in the derotated image of this film. This bounds fixed templates only, not probabilistic or equivariant predictors of the column statistics.

α ($^\circ$)	f_{train}	f_{hold}	Δ_{in}	Δ_{ho}	peak/mean
65	0.635	0.585	0.000	0.000	1.25
70	0.502	0.467	0.000	0.000	1.37
75	0.292	0.282	0.000	0.000	1.58
80	0.233	0.230	0.000	0.000	1.65
85	0.126	0.130	0.000	0.000	2.49
87	0.076	0.075	−0.0001	−0.0006	2.63
89	0.012	0.007	−0.0008	−0.0024	6.57

$S(\mathbf{q})$ accessible by small-angle or grazing-incidence X-ray scattering (SAXS/GISAXS) on nanostructured films [39]; the connection is subject to the form factor of the columns, their orientation distribution, and instrumental resolution, so $h_g(r)$ is not itself a directly measured pair-distribution function but a quantity Fourier-conjugate to the measurable scattering intensity, in the same sense as the pair-distribution function of a simple liquid is Fourier-conjugate to its static structure factor [40]. It quantifies how the correlation of column material decays with radial distance r from a reference point in the transverse plane. Unlike individual column positions (Section 4.1), $h_g(r; \alpha)$ is reproducible at the descriptor level across independent simulation runs at a given α , and varies smoothly with deposition angle. We verify this reproducibility directly using the three independent seeds available at $\alpha = 80^\circ$ and

85°, computing $h_g(r)$ separately for each seed. To avoid the optimistic bias of comparing each seed only against a mean that contains it, we report *independent* estimators: the mean pairwise (seed-to-seed) root-mean-square deviation is 0.0077 at 80° and 0.0090 at 85°, and the leave-one-seed-out deviation (each seed versus the mean of the other two) is 0.0066 and 0.0078, respectively (maximum pointwise standard deviation 0.010 at 80°, 0.007 at 85°). The self-inclusive deviation used in earlier drafts (0.002–0.007) understates the true variability. The CORRNIIF leave-one-angle-out error at these angles (RMSE $\approx$ 0.011, Table 4) is therefore comparable to, and about 1.2–1.7 $\times$ above, the independent seed-to-seed scatter: the curve is a genuinely reproducible ensemble descriptor, and the emulator reproduces it to within a small multiple of the simulator’s own run-to-run variability. The single-seed configurations at the remaining angles are used as-is; extending the three-seed test to the full angular range is identified as future work.

Definition. After bilinear derotation the corotating-frame field is a *continuous* occupancy field $J(\mathbf{r}) \in [0, 1]$ with mean $f = \langle J \rangle$ (equal to the binary filling fraction up to the interpolation error). We use the normalised covariance

$$C(r) = \langle J(\mathbf{x}) J(\mathbf{x} + r) \rangle, \quad (3)$$

$$h_g(r) = \frac{C(r) - f^2}{C(0) - f^2}, \quad h_g(0) = 1, \quad (4)$$

where $C(0) = \langle J^2 \rangle$ is read empirically from the zero-lag pixel. This continuous-covariance form (a standard normalised covariance) guarantees $h_g(0) = 1$ *without* requiring the binary identity $\langle J^2 \rangle = \langle J \rangle$, which the interpolated field does not satisfy: empirically $C(0)/f^2$ ranges from 1.35 (65°) to 59 (89°) and differs by 15–25% from the binary value $1/f$ at every angle, confirming that the normalisation must use the measured variance rather than $1/f$. Contrary to a naive expectation, $h_g(r)$ is *not* confined to $[0, 1]$: $C(r)$ may fall below f^2 where the arrangement is anticorrelated, so

1335 $h_g(r)$ takes small negative values at finite separation (down to ≈ -0.013 at $80-89^\circ$
1336 here). For an infinite statistically homogeneous system the correlation is expected to
1337 approach zero at large separation; the present periodic domain only samples $r \leq L/2$,
1338 so the long-range value is not a measured $h_g(\infty) = 0$ but a finite-box estimate. Equiv-
1339 alently, $h_g(r) = [C(r) - f^2]/[C(0) - f^2]$ is the connected correlation normalised by its
1340 own zero-lag value; its radial *shape* remains numerically sensitive as $f \rightarrow 0$ ($\alpha \rightarrow 90^\circ$),
1341 where both $C(r) - f^2$ and $C(0) - f^2$ become small. By construction $h_g(0) = 1$ for
1342 all f , so $h_g(0)$ encodes no filling-fraction information; the filling fraction is supplied
1343 separately from the binary field.

1344
1350 $h_g(r; \alpha)$ is computed in the corotating frame using bilinear derotation and the
1351 FFT-based circular autocorrelation, then radially averaged over the transverse plane
1352 (full pipeline in Section 2.6). Figure 10 shows the resulting curves for all seven con-
1353 figurations. *Derotation convention.* The curves shown, the cross-seed statistics, the
1354 leave-one-angle-out CORRNIFF retraining, and the corrected 72° baselines (Section 4.3)
1355 all use the periodic-wrap derotation and the continuous-covariance normalisation of
1356 Eqs. 3–4. Negative values of h_g are retained rather than clipped.

1357
1361 CORRNIFF *architecture* (see Section 2.6 for full specification). A compact NIF maps
1362 $(\alpha_{\text{norm}}, r_{\text{norm}}) \rightarrow h_g$, with a random Fourier encoder on r ($n_{\text{freq}} = 12$, $\sigma_r = 2$ [26]),
1363 three hidden layers (width 64, SiLU), and angle conditioning (FiLM) on α .

1364
1366 *LOOA validation.* In a seven-fold leave-one-angle-out (LOOA) experiment, COR-
1367 RNIFF is trained on six angles and used to predict $h_g(r; \alpha_{\text{held}})$. Figure 11 shows LOOA
1368 predictions against the reference curves. Results are summarised in Table 4.

1369
1371 The corrected LOOA results confirm that $h_g(r; \alpha)$ is a smoothly learnable function
1372 of α in the interior of the angle sweep. For the five interpolation folds, COR-
1373 RNIFF reaches mean $R^2 = 0.9923$ (median 0.9954, worst case 0.9768). Crucially, this
1374 smoothness is not unique to the neural emulator: on the same corrected folds, nearest-
1375 neighbour, linear, and PCHIP baselines also reproduce the held-out curves about
1376
1377
1378
1379
1380

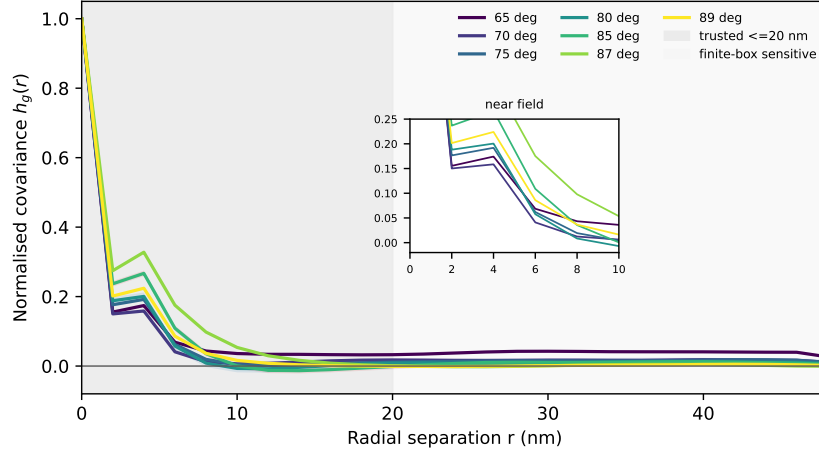


Fig. 10 The column-arrangement statistics change smoothly and systematically with deposition angle, unlike the individual column positions themselves, making this curve a promising reproducible target for interpolation across angles. Normalised two-point pair correlation $h_g(r; \alpha)$ in the corotating frame, for all seven deposition angles ($\alpha = 65^\circ$ – 89° , viridis colour scale; dense to sparse). This real-space descriptor is Fourier-conjugate to the structure factor $S(\mathbf{q})$ probed by SAXS/GISAXS: it quantifies how the correlation of column material decays with radial distance r , normalised to unity at $r = 0$. Curves differ systematically in amplitude and tail decay across α , providing the smooth signal for pair-correlation emulation; shaded uncertainty bands are shown where replicated corrected curves are available. All curves satisfy $h_g(0) = 1$ by construction and take small negative values at finite r where the arrangement is anticorrelated; the long-range tail is a finite-box ($r \leq L/2$) estimate, not a measured $h_g(\infty) = 0$.

equally well (Table 4). We therefore do not claim that CORRNIIF interpolates *more accurately* than these simpler methods—it does not. Its advantage is qualitative: it provides a single differentiable, continuous, multi-output representation $h_g(r; \alpha)$ that can be queried and differentiated at arbitrary (α, r) and extended to additional outputs, which is convenient for downstream gradient-based optimisation but not necessary for the present interpolation task, where a simpler method would suffice. The extrapolation folds are weaker: the CORRNIIF boundary mean is $R^2 = 0.9480$, with the lowest performance at $\alpha = 89^\circ$ ($R^2 = 0.9155$), the most extreme sparse-boundary case in the sweep. This underscores that no single interpolator dominates and that the choice should be driven by the downstream use rather than by held-out accuracy alone.

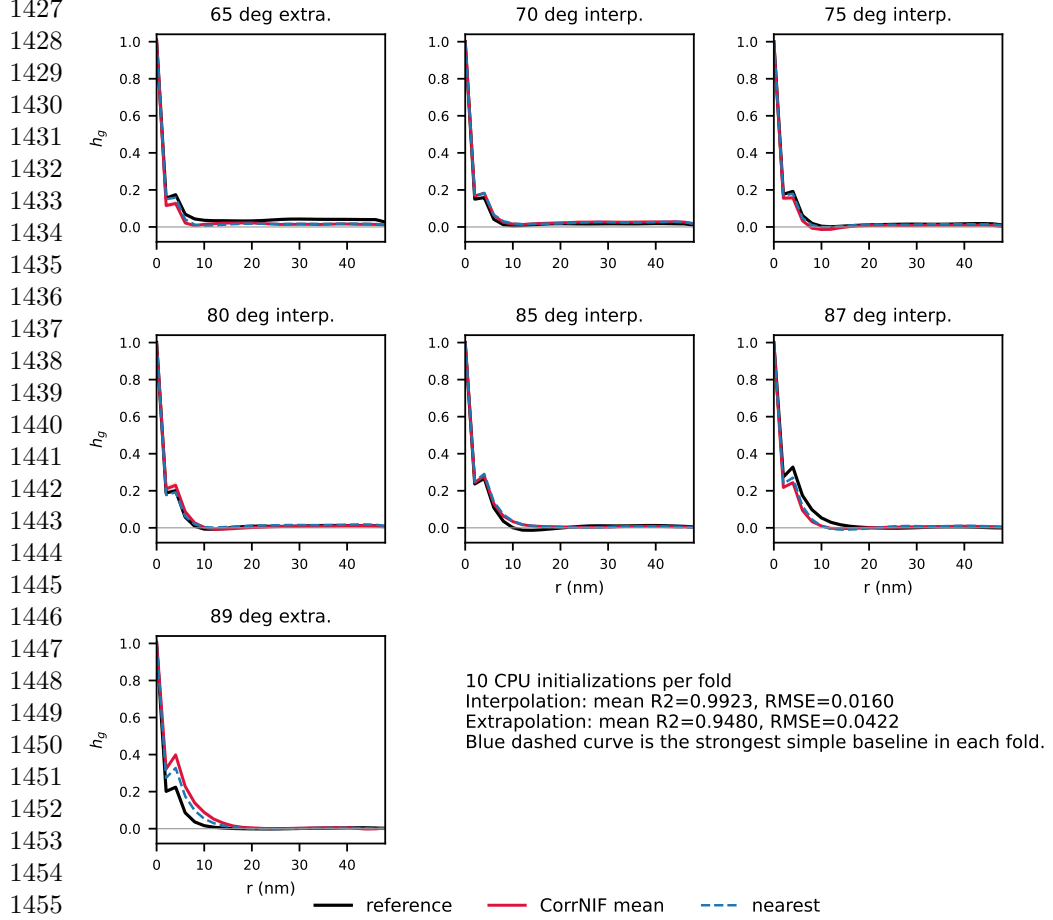


Fig. 11 The trained emulator successfully reconstructs the column-arrangement statistics at deposition angles it never saw during training, but so do much simpler interpolation methods, so its practical value lies in convenience rather than superior accuracy. Leave-one-angle-out CORRNIF predictions of $h_g(r; \alpha_{\text{held}})$. In each panel, the emulator was trained on the six remaining deposition angles and had no access to the held-out angle—analogueous to fitting an optical dispersion model to six incidence-angle spectra and predicting the seventh. Black solid lines: corrected reference curves computed from voxel data. Crimson solid lines and bands: mean and one-standard-deviation range of ten independently initialised CORRNIF trainings. Blue dashed lines: the strongest simple baseline for the corresponding fold, evaluated against the same corrected target. Panel labels identify interpolation and extrapolation folds. The emulator interpolates the interior folds accurately but does not outperform the strongest simple baselines on accuracy.

4.3 Prospective interpolation at $\alpha = 72^\circ$

As a prospective check, we query CORRNIF—trained on all seven existing configurations $\alpha \in \{65, 70, 75, 80, 85, 87, 89\}^\circ$ —at the deposition angle $\alpha = 72^\circ$, which is

Table 4 Leave-one-alpha-out (LOOA) performance of CORRNIIF versus interpolation baselines evaluated on the *same corrected* folds: nearest-neighbour (NN, copy the training curve at the closest α), linear blend (interpolate/extrapolate between the two nearest training angles), Gaussian-process regression (GP, squared-exponential kernel, fitted per radial bin), and monotone cubic (PCHIP) interpolation along α . R^2 : coefficient of determination on the held-out curve $h_g(r; \alpha_{\text{held}})$. The two sweep-boundary folds (65° , 89°) are *extrapolation*; the five interior folds are *interpolation* and are aggregated separately. CORRNIIF entries are the mean predictions from ten independent neural initialisations. CORRNIIF does not exceed the best simple baselines on accuracy: its interior mean R^2 (0.9923) is numerically comparable to linear and PCHIP (0.9928), while the extrapolation mean drops to 0.9480. The half-width error $|\Delta d_{\text{coi}}| = 2|\Delta r_{1/2}|$ is omitted because $r_{1/2} \approx 1.2 \text{ nm} \lesssim 1 \text{ vox}$ at all angles, below the 2 nm voxel resolution, so it is a sub-voxel curve-shape diagnostic rather than a physical column diameter.

$\alpha_{\text{held}} (^\circ)$	CORRNIIF	NN	Linear	GP	PCHIP
<i>Interior folds (interpolation)</i>					
70	0.9969	0.9872	0.9960	0.5602	0.9969
75	0.9954	0.9973	0.9994	0.8604	0.9987
80	0.9970	0.9989	0.9974	0.9283	0.9983
85	0.9953	0.9801	0.9932	0.9830	0.9899
87	0.9768	0.9810	0.9777	0.9852	0.9802
<i>Interior mean</i>	<i>0.9923</i>	<i>0.9889</i>	<i>0.9928</i>	<i>0.8634</i>	<i>0.9928</i>
<i>Interior median</i>	<i>0.9954</i>	<i>0.9872</i>	<i>0.9960</i>	<i>0.9283</i>	<i>0.9969</i>
<i>Boundary folds (extrapolation)</i>					
65	0.9805	0.9865	0.9865	-3.5859	0.9754
89	0.9155	0.9699	0.9699	0.8431	0.9267
<i>Boundary mean</i>	<i>0.9480</i>	<i>0.9782</i>	<i>0.9782</i>	<i>-1.3714</i>	<i>0.9511</i>
All-fold mean	0.9796	0.9858	0.9886	0.2249	0.9809

All R^2 values are coefficients of determination on the held-out curve. Interior interpolation is accurate and method-insensitive ($R^2 \approx 0.99$ for CorrNIIF, linear, and PCHIP); boundary extrapolation degrades, especially for CORRNIIF at 89° ($R^2 = 0.916$). The per-bin GP baseline is unstable for the corrected target under this small-angle training set, especially at the 65° boundary. Cubic spline interpolation (not shown) gives all-fold mean $R^2 = 0.957$.

genuinely absent from the training set. This point lies 40% of the way between the nearest training neighbours 70° and 75° , making it a mid-range interpolation rather than an extrapolation. Figure 12 shows this calibration design directly.

Figure 13 compares the corrected CORRNIIF prediction at 72° against the dedicated reference simulation and the strongest simple baseline, a linear blend $h_{\text{lin}}(r; 72^\circ) = 0.6 h_g(r; 70^\circ) + 0.4 h_g(r; 75^\circ)$. The CORRNIIF prediction interpolates

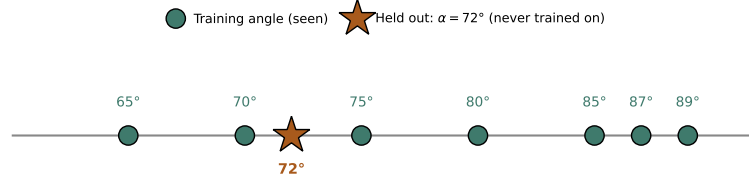


Fig. 12 The 72° test is a genuine blind interpolation, not a relabelled training point: this angle contributes nothing to CORRNIF’s parameters before the comparison in Figure 13 is made. CORRNIF’s training grid (green circles) versus the held-out prospective test angle (orange star, $\alpha = 72^\circ$), which is excluded from training and from all main-dataset descriptive analyses (Section 2.2).

smoothly between the two neighbouring training curves but is not more accurate than this linear baseline on the corrected target.

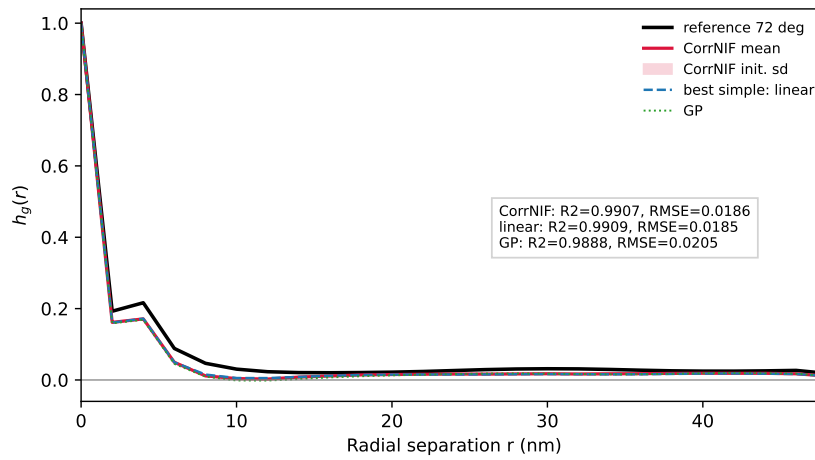


Fig. 13 A genuinely new deposition angle, never used for training and subsequently confirmed by an independent simulation, is predicted accurately by the emulator—though again no more accurately than a simple linear interpolation between the two nearest trained angles. CORRNIF prospective prediction versus dedicated reference simulation at $\alpha = 72^\circ$ (absent from the training set). Black solid: reference simulation $h_g(r; 72^\circ)$ ($h = 300$ nm, 68×10^6 atoms). Crimson solid and shaded band: mean and one-standard-deviation range of ten CORRNIF initialisations ($R^2 = 0.9907$ for the mean curve). Orange dotted: linear-blend baseline ($R^2 = 0.9909$). Green dash-dotted: GP baseline ($R^2 = 0.9888$). The comparison uses the same corrected periodic-wrap, continuous-covariance target for every method.

To validate this prospective prediction, a dedicated simulation at 72° was subsequently run under identical conditions (300 nm, 68×10^6 deposited atoms). Evaluated against this dedicated reference simulation on the corrected continuous-covariance target, CORRNIF gives $R^2 = 0.9907$ (RMSE 0.0186), while the linear blend gives

$R^2 = 0.9909$ (RMSE 0.0185), the best of the simple methods tested here. Nearest neighbour, PCHIP, cubic spline, and GP give $R^2 = 0.9882, 0.9902, 0.9904$, and 0.9888, respectively. We therefore do *not* claim that CORRNIF captures a non-linear α -dependence of the correlation tail that the simple interpolators cannot reproduce: for this single point they are numerically comparable. The result confirms that CORRNIF generalises to a genuinely unseen interpolation point, with no demonstrated accuracy advantage over simple interpolation.

4.4 Demonstrating the differentiability advantage:

gradient-based recovery of α

Sections 4.2–4.3 establish that CORRNIF’s only demonstrated advantage over simple, non-differentiable baselines is qualitative: because it is a continuous, differentiable function of α , it supports gradient-based queries that nearest-neighbour, linear, and PCHIP interpolation do not. We demonstrate this advantage directly with a minimal inverse-design experiment, rather than leaving it as an unexercised claim.

Task. Treat the dedicated-simulation-confirmed pair-correlation curve at $\alpha = 72^\circ$ (Section 4.3, genuinely absent from CORRNIF’s training set) as an observed target, and recover the deposition angle that produced it by gradient descent through the frozen, already-trained CORRNIF model—treating α itself as the only free parameter, with model weights held fixed. This mirrors a practical use case: given a desired or measured correlation shape, find the process angle that reproduces it without re-running the ballistic simulator or performing a discrete grid search.

Procedure and cross-check. Starting from five deliberately mismatched initial guesses spanning the training range ($\alpha_0 \in \{65, 70, 80, 85, 89\}^\circ$), we minimise the mean-squared error between CORRNIF’s output and the target curve using Adam (300 steps, learning rate 0.05 on the normalised angle). As an independent correctness check,

we also compute the loss on an exhaustive fine grid (0.005° resolution) over the full training range.

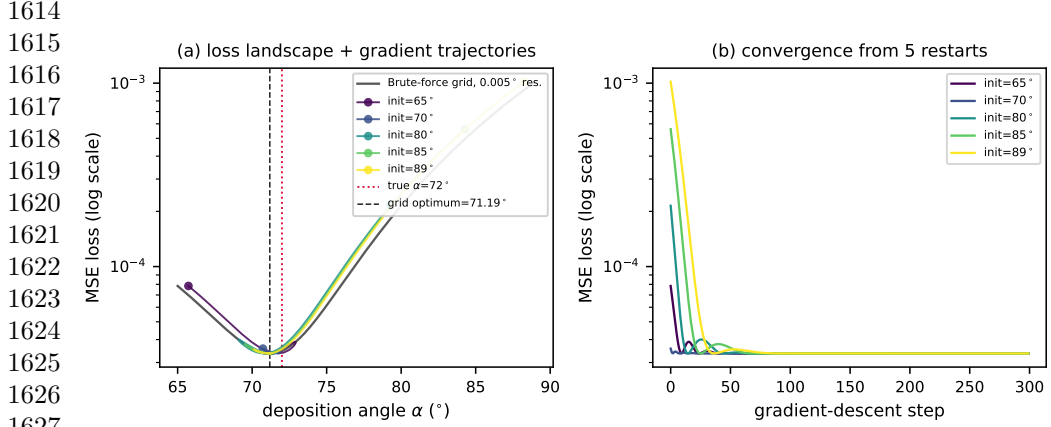


Fig. 14 Gradient descent through the frozen CORRIF model recovers the same angle from every tested starting point, and that angle matches an independent exhaustive grid search to within 0.002° —direct evidence that the differentiability claimed for CORRIF is usable, not just structurally present. (a) Brute-force loss landscape $L(\alpha)$ (grey) with the five gradient-descent trajectories overlaid, all converging to the same minimum regardless of starting point. (b) Per-restart loss-versus-iteration convergence, all five restarts reaching the same final loss within 300 steps.

Result. All five restarts converge to the identical recovered angle, $\alpha_{\text{rec}} = 71.183^\circ$ (standard deviation across restarts $< 10^{-4}^\circ$), regardless of starting point—including restarts initialised on the opposite side of the true angle from each other (Figure 14). This matches the independent brute-force grid optimum, 71.185° , to within 0.002° , confirming that gradient descent converges to the same optimum as exhaustive search rather than to an unrelated stationary point. Each restart converges in under 4 seconds on CPU (300 Adam steps); the one-time cost of training CORRIF itself is under one minute. We report this timing only to show the inverse query itself is cheap once trained, not as a claim that training plus inverse recovery is faster than a single ballistic simulation at a known angle.

We do *not* claim this recovers the true angle exactly: the recovered 71.183° differs from the actual deposition angle, $\alpha = 72^\circ$, by 0.82° . This residual is consistent with, not larger than, the interpolation error already reported for this same target curve in

Section 4.3 ($R^2 = 0.9907$, $\text{RMSE} = 0.0186$)—the inverse recovery is limited by the same finite-capacity, finite-training-grid interpolation error as the forward prediction, not by a failure of the optimisation procedure itself, which is independently validated by the exact grid-search agreement above. The demonstration therefore substantiates the differentiability advantage as claimed: a genuine gradient-based query is usable in practice, at a precision commensurate with, and no worse than, CORRNIIF’s already established forward-interpolation accuracy.

4.5 Effective optical properties: a loss-aware Maxwell-Garnett screening illustration

This subsection addresses **Q4**. We compute effective optical constants from the filling fractions $f(\alpha)$ using the aligned-cylinder Maxwell-Garnett (MG) effective-medium approximation [27] (see Section 2.6 for the model setup and its limitations). As a sensor-design hypothesis-generation illustration only, we take tungsten columns ($\varepsilon_{\text{W}} = 4.41 + 19.60i$ at $\lambda \approx 500$ nm, i.e. $n = 3.5$, $k = 2.8$ [28]) in vacuum, with structural $f(\alpha)$ extracted from the Cu GLAD simulations. This screening calculation uses the $R = 0.128$ nm (Cordero covalent-radius) campaign rather than the $R = 0.1$ nm campaign used for the void-fraction, reproducibility, and PCA results elsewhere in this paper (Table 1); the two radii give systematically slightly different filling fractions (void fraction ~ 0.3 – 0.5 percentage points lower at $R = 0.128$ nm), which does not change any qualitative conclusion drawn here but should not be conflated with the primary-campaign numbers reported in Section 3.1. Two further caveats govern the interpretation of this calculation. First, the filling fraction is definition-dependent: the bead-sphere solid fraction $1 - P$ and the thresholded voxel occupancy $f_{\rho>0.5}$ differ by up to a factor of four at low α (e.g. 0.115 vs 0.492 at 65° ; Table 5), and the thresholded occupancy is itself sensitive to the Gaussian voxelisation kernel and the 0.5 threshold rather than being a physical material volume fraction. Second, the columns are

strongly absorbing, so the extraordinary polarisation carries a large extinction k_e that must be reported alongside Δn . We therefore present the optical results as an illustrative, definition- and material-dependent screening calculation, not a quantitative design prediction, and report both filling-fraction conventions.

The ordinary index n_o (transverse polarisation, cylindrical inclusion geometry) is obtained from:

$$\varepsilon_o = \varepsilon_h \frac{(\varepsilon_c + \varepsilon_h) + f(\varepsilon_c - \varepsilon_h)}{(\varepsilon_c + \varepsilon_h) - f(\varepsilon_c - \varepsilon_h)}, \quad (5)$$

and the extraordinary index n_e (longitudinal, Wiener parallel limit) from:

$$\varepsilon_e = f \varepsilon_c + (1 - f) \varepsilon_h, \quad (6)$$

where ε_c is the column dielectric function and $\varepsilon_h = 1$ (vacuum host). The form birefringence is $\Delta n = |n_e - n_o|$.

Figure 15 and Table 6 give the results.

Table 5 Filling-fraction reconciliation. P : bead-sphere void fraction (Eq. 1); $1 - P$: bead-sphere solid fraction; $f_{\rho>0.5}$: thresholded voxel occupancy used in the optical calculation. The two solid-fraction measures differ by up to a factor of four and are not interchangeable; $f_{\rho>0.5}$ depends on the voxelisation kernel and threshold and is not a physical material volume fraction.

α ($^\circ$)	P (%)	$1 - P$	$f_{\rho>0.5}$	$f_{\rho>0.5}/(1 - P)$
65	88.5	0.115	0.492	4.3
70	89.5	0.105	0.409	3.9
75	90.9	0.091	0.348	3.8
80	92.7	0.073	0.197	2.7
85	95.3	0.047	0.113	2.4
87	96.6	0.034	0.070	2.0
89	98.1	0.019	0.011	0.6

For the thresholded-occupancy filling fraction, the real-index form birefringence is largest near $\alpha = 65^\circ$ ($\Delta n = 0.839$). This maximum should not be read as a usable

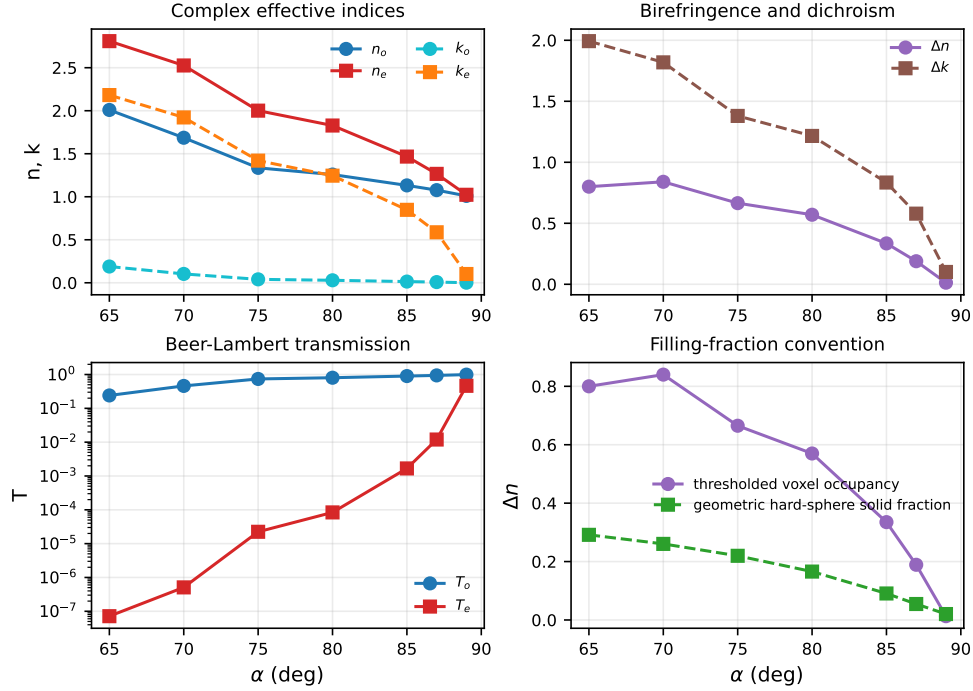


Fig. 15 A large apparent form birefringence at moderate obliquity is accompanied by near-total absorption of the extraordinary polarisation, and the apparent optimum angle is itself an artefact of how the solid filling fraction is defined—so this screening illustrates the morphology-to-optics pathway without identifying a usable polarimetric design window. Loss-aware aligned-cylinder Maxwell-Garnett optical screening of a W-column film versus α (illustrative material substitution; see Section 2.6). (a) Full complex effective indices for the thresholded-occupancy filling fraction: ordinary n_o and extraordinary n_e (solid), ordinary k_o and extraordinary k_e (dashed). The extraordinary extinction k_e is of order unity. (b) Form birefringence $\Delta n = |n_e - n_o|$ and dichroism $\Delta k = |k_e - k_o|$; the dichroism exceeds the birefringence across the sweep. (c) Beer-Lambert intensity transmission of the two eigenpolarisations through the 300 nm film (log scale): the extraordinary transmission T_e stays far below 1% throughout the high- Δn range, i.e. the extraordinary eigenstate is essentially fully absorbed. (d) Δn computed with the two filling-fraction conventions: the apparent maximum near 65° (thresholded occupancy $f_{\rho>0.5}$, $\Delta n = 0.839$) disappears for the geometric hard-sphere solid fraction $1 - P$, which decays monotonically. No polarimetric design window is claimed; the authoritative tabulated values are in Tables 5 and 6.

design optimum, for two reasons. First, it is an artefact of the filling-fraction definition: recomputing the same MG model with the geometric hard-sphere solid fraction $1 - P$ (Table 5) gives a birefringence that is roughly three times smaller and *monotonically decreasing* with α ($\Delta n \approx 0.30$ at 65° falling to 0.03 at 89°), with no interior peak. The location and magnitude of the “optimum” are therefore not robust to how the solid fraction is defined. Second, and independently of the definition, the

Table 6 Aligned-cylinder Maxwell-Garnett effective optical constants [27] for a high-index columnar film ($\epsilon_W = 4.41 + 19.60i$, $\lambda \approx 500$ nm [28], vacuum host), using the thresholded voxel occupancy $f = f_{\rho > 0.5}$. Both polarisations are now reported as full complex indices. n_o, k_o : ordinary index and extinction; n_e, k_e : extraordinary index and extinction; $\Delta n = |n_e - n_o|$; T_e : extraordinary intensity transmission through the 300 nm film (Beer–Lambert screening, normal incidence, interfaces neglected). The large k_e and near-zero T_e across the high- Δn range show that the extraordinary eigenstate is essentially fully absorbed; the figures are an illustrative screening calculation only (see Table 5 for the filling-fraction caveat).

α (°)	f	n_o	k_o	n_e	k_e	Δn	Δk	T_e
65	0.492	1.679	0.102	2.518	1.914	0.839	1.81	5×10^{-7}
70	0.409	1.522	0.070	2.320	1.728	0.798	1.66	2×10^{-6}
75	0.348	1.421	0.053	2.161	1.576	0.740	1.52	7×10^{-6}
80	0.197	1.214	0.024	1.713	1.125	0.500	1.10	2×10^{-4}
85	0.113	1.116	0.012	1.412	0.781	0.296	0.77	3×10^{-3}
87	0.070	1.070	0.007	1.241	0.551	0.171	0.54	2×10^{-2}
89	0.011	1.011	0.001	1.025	0.110	0.014	0.11	0.44

extraordinary eigenstate is almost completely absorbed across the high- Δn range: the extraordinary extinction is $k_e \approx 1.9$ at 70° , giving a Beer–Lambert intensity transmission of $T_e \sim 10^{-6}$ through the 300 nm film (Table 6). A large Δn accompanied by near-total absorption of one polarisation does not support a polarimetric device, so we do *not* claim that any angular window is suitable for guided-mode-resonance or other polarimetric biosensors; establishing device suitability would require a loss-aware figure of merit and a device-level transmission or resonance calculation that are beyond the present effective-medium screening. With these caveats, the qualitative angular trend—stronger form birefringence at moderate obliquity, decaying as $f \rightarrow 0$ toward 90° —is consistent with angle-dependent optical anisotropy in GLAD copper-oxide films. In Cu_2O zigzag GLAD films grown by thermal vacuum evaporation at oblique incidence and oxidised by post-deposition annealing, a birefringence maximum of $\Delta n = 0.106$ at a deposition angle of $\gamma = 60^\circ$ (using the source studies’ own symbol γ for the same deposition-angle quantity denoted α throughout this paper; optical constants $n = 2.42$ – 2.62) has been reported [11]. A subsequent CuO_x evaporation series reported the same angular trend: $\Delta n = 0.20$ at $\gamma = 60^\circ$ declining to $\Delta n = 0.04$ at $\gamma = 85^\circ$ for the Cu_2O phase [41]. These two studies share authorship

and an experimental lineage, so they are corroborating rather than fully independent; they indicate that GLAD optical anisotropy in this oxide system is largest near 60° – 70° and decays at higher obliquity, consistent with the present trend. The much larger absolute magnitude obtained here for tungsten ($n = 3.5$, $k = 2.8$) reflects its far stronger dielectric contrast, but in view of the absorption it is not a directly realisable optical advantage and remains an illustrative material substitution pending direct W-GLAD ellipsometry. Tungsten, rather than the project’s own $\text{Cu}_2\text{O}/\text{CuO}_x$ optical constants (already used quantitatively above via [11, 41]), was chosen deliberately as a generic high-index, strongly-absorbing metal so that the Maxwell-Garnett screening in this subsection isolates the purely geometric form-birefringence mechanism from any particular material’s band-structure-specific optical response; the oxide-specific behaviour of $\text{Cu}_2\text{O}/\text{CuO}_x$ is instead addressed qualitatively via the direct literature comparison above.

5 Discussion

5.1 Scope of the ballistic model

The results demonstrate that the ballistic GLAD simulator reproduces the expected *monotonic trend* of increasing void fraction with deposition angle across the full simulated angle range, but *quantitatively overestimates* the void fraction relative to published optical void fractions of other GLAD metals—a cross-material, cross-method discrepancy, not a calibrated offset. The model is most reliable for relative comparisons within the same simulation campaign (e.g., comparing $\alpha = 75^\circ$ to $\alpha = 85^\circ$) and less reliable for absolute predictions of experimentally measurable quantities without calibration. A Monte Carlo growth-dynamics study spanning multiple sputtered metals and oxides found that removing hyperthermal surface-relaxation processes from an otherwise similar model reproduces the thin, tilted, lower-density morphology characteristic of thermal evaporation [42], supporting the classification of the present

1887 purely ballistic, no-relaxation model as representative of the evaporation-like deposi-
1888 tion regime rather than the densified sputtering regime; this is consistent with, but
1889 does not resolve, the absence of a calibrated surface-relaxation term noted above.
1890
1891

1892

1893 **5.2 Reproducibility as a model quality criterion**

1894

1895 The sub-0.25% CV in atom areal density across three independent seeds demon-
1896 strates that the stochastic element of the ballistic simulation (random azimuthal flux,
1897 Poisson-distributed arrival positions) does not introduce significant variability in the
1898 macroscopic void fraction at the simulated scale. This is consistent with the statistical-
1899 mechanics argument that large- N depositions converge to ensemble-average behaviour.
1900 We stress the limited scope of this conclusion: for the global void fraction at 80°
1901 and 85° one realization appears sufficient at the reported precision, but this is not
1902 established for other angles or for higher-order descriptors (the pair-correlation curve,
1903 for example, retains a measurable seed-to-seed scatter, Section 4.2). Extending the
1904 three-seed test across the full angular range remains future work.
1905
1906
1907
1908
1909
1910
1911

1912

1913 **5.3 Depth gradients: implications for property predictions**

1914

1915 The dense substrate layer revealed in the Z-profiles (Section 3.2) would affect any
1916 optical model that integrates the refractive index through the film thickness. Effective-
1917 medium calculations that assume a spatially uniform void fraction (as typically
1918 used in [10]) would under-estimate the substrate-layer contribution. This observation
1919 motivates depth-graded effective-medium models as a more rigorous alternative.
1920
1921
1922
1923

1924

1925 **5.4 Limitations and future work**

1926

1927 This study is subject to the following limitations:

1928

- 1929 1. No surface diffusion: the model cannot reproduce experimental void fractions
1930 without calibration.
1931
1932

2. Cu-specific validation: no optical void-fraction sweep versus α is available in the published literature specifically for Cu GLAD films under conditions comparable to the present simulations; direct quantitative comparison between the simulation trend and a Cu experimental trend therefore remains future work.	1933 1934 1935 1936 1937 1938 1939
3. Single helical pitch: all simulations use pitch = 150 nm; pitch dependence of void fraction is not explored here.	1940 1941 1942
4. Voxelisation artefact at $\alpha = 60^\circ$: excluded from voxel analyses; root cause (kernel bandwidth mismatch) is flagged for repair in future pipeline updates.	1943 1944 1945
5. Lateral box size is not fully converged. A supplementary finite-size study (Appendix A), carried out at reduced film heights ($h = 25\text{--}100$ nm for computational tractability), shows the deposited areal density continues to decrease with lateral box size L with no plateau up to $L = 350\text{--}450$ nm, and that the run-to-run (seed) scatter <i>grows</i> with L rather than self-averaging away. Consequently the <i>absolute magnitudes</i> reported here (void-fraction endpoints, PCA loadings, CORRNIIF R^2 , Maxwell-Garnett Δn) should be read as characteristic of the 100 nm-box, $h = 300$ nm convention used throughout this study rather than as converged infinite-box values; a box exactly equal to the pitch ($L = 150$ nm) additionally shows a commensurability artefact and should be avoided. Crucially, the paper's two central findings are <i>relational</i> , <i>not absolute</i> , and are therefore not sensitive to this magnitude caveat: (i) the oracle result that absolute column positions carry no realization-invariant signal is a statement about independence across seeds, which no change of box size can manufacture; and (ii) the smooth angular interpolability of the pair correlation (CORRNIIF) is a statement about continuity in α at fixed convention. We nonetheless flag that the <i>quantitative</i> void-fraction values and the exact magnitude of the angular trend remain provisional pending an angle-resolved, full-height convergence study (future work).	1946 1947 1948 1949 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977 1978

Future work should address (i) addition of a phenomenological diffusion term to reduce the void-fraction overestimate relative to experiment, (ii) experimental validation with Cu GLAD films over the $\alpha = 70\text{--}89^\circ$ range, (iii) a lateral box-size convergence study specifically for the pair-correlation function used by the NIF/CorrNIF pipeline (Section 4.2), to confirm the short-range correlation is free of finite-size artefacts – this remains open (the bulk atom-density convergence check described in Section 5.4 above tests a different, coarser observable and does not by itself establish correlation-function convergence), and (iv) extension of the parameter sweep to the pitch dimension.

5.5 NIF scope: realization dependence versus reproducible ensemble statistics

Section 4 establishes a practical hierarchy of what the voxel data can and cannot support. The oracle experiment (Table 3) shows empirically that, for this simulator’s pooled corotating protocol, the best fixed spatial template (the mean derotated occupancy map) carries no information beyond the bulk filling fraction, so a coordinate-conditioned model of the form $f(x, y, z, \alpha) \rightarrow I$ has no realization-invariant target to recover: column positions are set independently in each run and do not transfer between independent realizations. Because the oracle measures the best fixed mean-template baseline rather than a particular trained network, this is stronger evidence than the flat training loss alone—it indicates that the absence of a recoverable absolute-position signal under a fixed template is a property of the stochastic growth protocol, not merely of one architecture or optimiser (it does not bound probabilistic or equivariant predictors of the statistics). This is not a claim that a voxel field is unrepresentable: our controlled single-realization overfit (Section 4.1) shows that the coordinate network memorises a deterministic transverse slice and a sub-volume of one field essentially perfectly (IoU = 1.00 and 0.997, BCE $\rightarrow$ 0). The limitation is specifically one of *transfer*—an absolute occupancy map fitted to one realization does

not generalise to an independently nucleated run, because the column positions are re-randomised between runs.

By contrast, the two-point pair correlation $h_g(r; \alpha)$ encodes the *ensemble statistics* of the column arrangement, which are approximately reproducible under the tested seeds and vary smoothly with α in the present dataset. The corrected CORRNIFF LOOA results (interior mean $R^2 = 0.9923$, extrapolation mean $R^2 = 0.9480$) show that this quantity is learnable in the interior of the sweep but materially less secure at the sparse-angle boundary. The prospective validation at $\alpha = 72^\circ$ ($R^2 = 0.9907$ against a dedicated simulation) further confirms that the NIF generalises beyond the leave-one-out setting to a genuinely unseen interpolation point, without demonstrating an accuracy advantage over linear interpolation ($R^2 = 0.9909$). The practical implication is that statistical descriptors—filling fraction, pair correlation half-width, column density—are the correct targets for data-driven GLAD modelling, not individual voxel occupancies.

For downstream optical screening, this finding has practical value. The illustrative form-birefringence estimates of Section 4.5 were derived entirely from the structural $f(\alpha)$ of the ballistic simulations, without additional experimental input; as discussed there, their absolute magnitude and the apparent angular optimum are not quantitatively defensible (filling-fraction ambiguity and strong extraordinary absorption), so they serve only to exercise the morphology-to-optics pathway rather than to define a usable sensor window. A dedicated companion study [38] investigates the voxel-occupancy learnability question in depth, including conditioning-based remedies and a multi-seed enlargement test. Once CORRNIFF is trained on a small set of simulation runs, it provides a continuous emulator for the correlation *shape* $h_g(r; \alpha)$ at any intermediate α . The filling fraction $f(\alpha)$ required for the Maxwell-Garnett step is supplied separately (by interpolating the measured $f(\alpha)$ values directly, as $h_g(0) = 1$ is independent of f and carries no filling-fraction information), enabling

rapid effective-medium parameter sweeps without additional simulation cost. This positions the statistical-morphology NIF as a candidate differentiable component for future morphology-screening workflows: deposition angle can be swept analytically through the emulator rather than through repeated full simulations, reducing the cost of an angle sweep from N_α simulations to a single trained-network evaluation. We make no closed-loop or device-optimisation claim; the optical step remains illustrative and experimentally uncalibrated.

5.6 Effective-medium birefringence estimate in context

The Maxwell-Garnett calculation (Section 4.5) is a loss-aware effective-medium screening illustration, not an experimental validation or a device design: it uses measured W optical constants [28] with geometry-derived filling fractions in an aligned-cylinder approximation that omits the rotating optic axis and chirality of the true helical film. Its quantitative outputs are limited by two factors established above: the form birefringence depends strongly on which solid-fraction definition is used (the interior Δn maximum vanishes for the geometric hard-sphere fraction), and the extraordinary eigenstate is almost fully absorbed ($k_e \sim 1$, $T_e \sim 10^{-6}$ through 300 nm). We therefore retain only the qualitative angular trend—larger form birefringence at moderate obliquity, decaying toward 90° —which is consistent with the Cu_2O GLAD measurements of Chaffar Akkari et al. [11, 41] (these two share authorship and lineage and are corroborating rather than fully independent). We make no claim about the absolute W-column magnitude or about polarimetric device suitability; genuine validation requires generalised (anisotropic) W-GLAD ellipsometry and a device-level optical model, both identified as primary future experiments. Furthermore, the transfer of filling fractions from Cu ballistic simulations to W, Mo, or Co columns assumes comparable columnar geometry; in practice, material-specific surface diffusion, sticking coefficients, and thermally activated adatom mobility will shift the true filling fractions

and must be calibrated experimentally against material-specific GLAD depositions before quantitative predictions can be made.

The sensor-application chain is further grounded by direct evidence of Cu-oxide GLAD as a gas-sensing platform: Benamor et al. [12] demonstrated that Cu_2O GLAD films on glass exhibit surface roughness R_a increasing from 4.84 to 8.22 nm with deposition angle, directly confirming the $\alpha \rightarrow \text{morphology} \rightarrow \text{sensor response}$ relation that the present morphology workflow targets. Together, the ballistic simulation (with its characterised cross-material discrepancy), the CORRINF morphology emulator, and the illustrative MG calculation form a simulation-only workflow for ranking helical GLAD geometries by ensemble morphology and screening their *candidate* optical behaviour, with the optical step explicitly flagged as illustrative pending material- and device-specific validation.

6 Conclusion

We have characterised the scope of a bead-sphere ballistic GLAD simulator across twelve helical Cu configurations spanning $\alpha = 60\text{--}89^\circ$ and demonstrated that its morphological outputs can be partitioned into two distinct predictability classes. We close by answering, in order, the four questions (**Q1–Q4**) posed in Section 1.

Answering Q1 (void-fraction trend, reproducibility, and comparison with experiment): the bead-sphere void fraction P increases monotonically from 88.1% to 98.5% over the tested angle range with sub-0.25% run-to-run variability at the 100 nm-box, $h = 300$ nm convention (Appendix A shows this reproducibility is convention-dependent, degrading at larger lateral box size), confirming that the bulk void fraction is well-defined and reproducible at the descriptor level within that convention. Against published optical void fractions of other GLAD metals at $\alpha = 85^\circ$, the simulator’s bead-sphere void fraction is +20.7 pp higher; this is a cross-material, cross-method

2163 discrepancy—attributable to the absence of surface diffusion, the differing measure-
2164 ment convention, and the different materials—not a calibrated offset, and Cu-specific
2165 measurements would be needed to establish a true calibration.
2166

2167 **Answering Q2** (depth dependence and low-dimensional morphology summary):
2168 the depth-resolved void-fraction profiles show a denser substrate layer ($z < 30$ nm)
2169 giving way to a shadowing-dominated bulk regime whose void fraction increases with
2170 α , a gradient that matters for any optical model that integrates the refractive index
2171 through the film thickness. An exploratory principal-component summary of three
2172 non-redundant morphology descriptors captures 75% of their variance in two com-
2173 ponents, with deposition angle as the dominant controlled source of variation; the
2174 higher-order descriptors (anisotropy, width gradient) show visible seed-to-seed scat-
2175 ter, so we read this as an exploratory descriptor-space summary rather than evidence
2176 of a smooth low-dimensional manifold.
2177

2178 **Answering Q3** (learnability of morphology by a neural implicit field): individual
2179 absolute column-position maps are not a stable target across independent realiza-
2180 tions under the stochastic-nucleation protocol of this simulator. A corotating-frame
2181 oracle experiment shows empirically that the best fixed mean-template baseline gives
2182 zero IoU gain ($\Delta = 0.000$) over the all-positive baseline ($\text{IoU} = f$) at all seven
2183 tested angles, so absolute column positions carry no realization-invariant signal that a
2184 coordinate-conditioned model could recover. The normalised two-point pair correlation
2185 $h_g(r; \alpha)$ sidesteps this limitation by encoding ensemble statistics rather than individ-
2186 ual absolute column positions, and is approximately reproducible within the tested
2187 three-seed cases at 80° and 85° (independent pairwise seed-to-seed RMSE 0.008–0.009,
2188 leave-one-seed-out 0.007–0.008). A compact parametric emulator (CORRNIF) trained
2189 on six deposition angles achieves an interior leave-one-angle-out mean $R^2 = 0.9923$
2190 and an extrapolation mean $R^2 = 0.9480$, comparable to the strongest simple base-
2191 lines on interpolation folds but weaker at the 89° boundary: the emulator does not
2192
2193
2194
2195
2196
2197
2198
2199
2200
2201
2202
2203
2204
2205
2206
2207
2208

improve interpolation *accuracy* over these simpler methods, and its value is instead the differentiable, continuous, multi-output representation it provides for downstream optimisation. A prospective test at $\alpha = 72^\circ$ —genuinely absent from the training set and subsequently confirmed by a dedicated simulation—yields $R^2 = 0.9907$ for CORRNIF; a simple linear interpolation between the neighbouring angles is marginally higher on the same corrected target ($R^2 = 0.9909$). This demonstrates that CORRNIF generalises beyond the leave-one-angle-out setting without a demonstrated accuracy advantage over simple interpolation.

Answering Q4 (optical-anisotropy implications of the morphology): an illustrative aligned-cylinder Maxwell-Garnett calculation maps the extracted filling fractions to candidate optical constants. We do not draw a quantitative design window from it: the apparent real-index birefringence maximum ($\Delta n = 0.839$ at 65° for the thresholded-occupancy fraction) disappears when the geometric hard-sphere fraction is used, and the extraordinary eigenstate is almost fully absorbed ($k_e \sim 1$, $T_e \sim 10^{-6}$ through 300 nm). The retained result is only the qualitative angular trend, which is consistent with the Cu_2O and CuO_x GLAD measurements of Chaffar Akkari et al. [11, 41] (a single shared experimental lineage, not independent confirmations). Together, the ballistic simulator (with its characterised cross-material discrepancy), the CORRNIF statistical-morphology emulator, and the illustrative effective-medium step form a simulation-only workflow for ranking helical GLAD geometries by ensemble morphology; quantitative optical design and material calibration are left to dedicated experiments. The principal identified limitations—absence of surface diffusion, restriction to Cu hard-sphere geometry, and the single helical pitch studied—define clear targets for future extension: addition of a phenomenological diffusion correction, experimental validation with Cu GLAD films, and exploration of pitch-dependent morphology effects.

2255 **Acknowledgements.** The authors acknowledge the institutional and scientific sup-
2256 port of the Laboratoire de Photovoltaïque et Matériaux Semi-conducteurs (LPMS),
2257 École Nationale d'Ingénieurs de Tunis (ENIT), Université de Tunis El Manar, and the
2258 Institut des NanoSciences de Paris (INSP), Sorbonne Université, within the framework
2260 of the PHC-Utique project.
2261
2262

2263

2264

2265 **Declarations**

2266

2267 • **Funding:** No specific financial support statement is declared for this manuscript
2268 because no verified project number, contract number, or funding-agency wording
2269 was found in the local project records.
2270

2271 • **Conflict of interest:** The authors declare no competing interests.
2272

2273

2274 • **Ethics approval and consent to participate:** Not applicable.
2275

2276 • **Consent for publication:** Not applicable.
2277

2278 • **Data availability:** The simulation data and analysis code supporting the findings
2279 of this study are available from the corresponding author upon reasonable request.
2280

2281 • **Materials availability:** Not applicable.
2282

2283 • **Code availability:** The analysis, corrected pair-correlation, figure-generation, and
2284 CORRNIF training code supporting the findings of this study are available from the
2285 corresponding author upon reasonable request.
2286

2287 • **Author contribution:** O.B.H.A. designed and performed all simulations, devel-
2288 oped the NIF training pipeline, and wrote the manuscript. F.C.A., B.G., and N.M.
2289 supervised the work and critically revised the manuscript.
2290

2291

2292

2293 **References**

2294

2295
2296 [1] Robbie, K., Brett, M.J., Lakhtakia, A.: Chiral sculptured thin films. Nature
2297 **384**(6610), 616 (1996) <https://doi.org/10.1038/384616a0>
2298
2299

2300

- [2] Hawkeye, M.M., Taschuk, M.T., Brett, M.J.: Glancing Angle Deposition of Thin Films: Engineering the Nanoscale. Wiley, Chichester, UK (2014)
- [3] Barranco, A., Borrás, A., Gonzalez-Elipe, A.R., Palmero, A.: Perspectives on oblique angle deposition of thin films: From fundamentals to devices. Progress in Materials Science **76**, 59–153 (2016) <https://doi.org/10.1016/j.pmatsci.2015.06.003>
- [4] Hawkeye, M.M., Brett, M.J.: Glancing angle deposition: Fabrication, properties, and applications of micro- and nanostructured thin films. Journal of Vacuum Science & Technology A **25**(5), 1317–1335 (2007) <https://doi.org/10.1116/1.2764082>
- [5] Brett, M.J., Hawkeye, M.M.: New materials at a glance. Science **319**(5867), 1192–1193 (2008) <https://doi.org/10.1126/science.1153910>
- [6] Lakhtakia, A., Messier, R.: Sculptured Thin Films: Nanoengineered Morphology and Optics, p. 299. SPIE Press, Bellingham, WA (2005)
- [7] Messier, R., Venugopal, V.C., Sunal, P.D.: Origin and evolution of sculptured thin films. Journal of Vacuum Science & Technology A **18**(4), 1538–1545 (2000) <https://doi.org/10.1116/1.582381>
- [8] Steele, J.J., Brett, M.J.: Nanostructure engineering in porous columnar thin films: Recent advances. Journal of Materials Science: Materials in Electronics **18**(4), 367–379 (2007) <https://doi.org/10.1007/s10854-006-9049-8>
- [9] Robbie, K., Brett, M.J.: Sculptured thin films and glancing angle deposition. Journal of Vacuum Science & Technology A **15**(3), 1460–1465 (1997)
- [10] Fischer, D., Schubert, M.: Generalized ellipsometry determinations of sculptured

thin films. In: Losurdo, M., Hingerl, K. (eds.) Ellipsometry at the Nanoscale, pp. 563–596. Springer, Berlin, Heidelberg (2013)

[11] Chaffar Akkari, F., Ben Jbara, H., Abdelkader, D., Gallas, B., Kanzari, M.: Effect of angle deposition γ on the structural, optical and electrical properties of copper oxide zigzag ($+\gamma$, $-\gamma$) nanostructures elaborated by glancing angle deposition. Thin Solid Films **657**, 61–69 (2018) <https://doi.org/10.1016/j.tsf.2018.05.006>

[12] Benamor, M., Gannouni, J., Chaffar Akkari, F.: Elaboration and characterization of Cu₂O thin films via glancing angle deposition for high-performance gas sensing of ethanol and methanol at room temperature. Materials Chemistry and Physics **349**, 131776 (2026) <https://doi.org/10.1016/j.matchemphys.2025.131776>

[13] Karabacak, T., Singh, J.P., Zhao, Y.-P., Wang, G.-C., Lu, T.-M.: Scaling during shadowing growth of isolated nanocolumns. Physical Review B **68**, 125408 (2003) <https://doi.org/10.1103/PhysRevB.68.125408>

[14] Nieuwenhuizen, J.M., Haanstra, H.B.: Microfractography of thin films. Philips Technical Review **27**, 87–91 (1966)

[15] Tait, R.N., Smy, T., Brett, M.J.: Modelling and characterization of columnar growth in evaporated films. Thin Solid Films **226**, 196–201 (1993) [https://doi.org/10.1016/0040-6090\(93\)90378-3](https://doi.org/10.1016/0040-6090(93)90378-3)

[16] Lichter, S., Chen, J.: Model for columnar microstructure of thin solid films. Physical Review Letters **56**(13), 1396–1399 (1986) <https://doi.org/10.1103/PhysRevLett.56.1396>

[17] Smy, T., Vick, D., Brett, M.J., Dew, S.K., Wu, A.T., Sit, J.C., Harris, K.D.: Three-dimensional simulation of film microstructure produced by glancing angle deposition. Journal of Vacuum Science & Technology A **18**(5), 2507–2512 (2000)

<https://doi.org/10.1116/1.1286394> 2393
2394

[18] Suzuki, M., Taga, Y.: Numerical study of the effective surface area of obliquely 2395
deposited thin films. *Journal of Applied Physics* **90**(11), 5599–5605 (2001) [https:](https://doi.org/10.1063/1.1415534) 2397
[//doi.org/10.1063/1.1415534](https://doi.org/10.1063/1.1415534) 2398
2399
2400

[19] Kulkarni, O., Qu, C., Walsh, K., McNamara, S.: A 3D Simulation of Glancing 2401
Angle Deposition. University of Utah / University of Louisville, Department of 2402
Electrical and Computer Engineering; manuscript PDF dated 2022, no DOI or 2403
proceedings venue confirmed (2022) 2404
2405
2406
2407
2408

[20] Meakin, P., Jullien, R.: Restructuring effects in the rain model for random depo- 2409
sition. *Journal de Physique* **48**(10), 1651–1662 (1987) [https://doi.org/10.1051/](https://doi.org/10.1051/jphys:0198700480100165100) 2410
[jphys:0198700480100165100](https://doi.org/10.1051/jphys:0198700480100165100) 2411
2412
2413
2414

[21] Thornton, J.A.: Influence of apparatus geometry and deposition conditions on the 2415
structure and topography of thick sputtered coatings. *Journal of Vacuum Science* 2416
& Technology **11**(4), 666–670 (1974) 2417
2418
2419
2420

[22] Sitzmann, V., Martel, J.N.P., Bergman, A.W., Lindell, D.B., Wetzstein, G.: 2421
Implicit neural representations with periodic activation functions. In: *Advances* 2422
in Neural Information Processing Systems (NeurIPS), vol. 33 (2020) 2423
2424
2425
2426

[23] Park, J.J., Florence, P., Straub, J., Newcombe, R., Lovegrove, S.: DeepSDF: 2427
Learning continuous signed distance functions for shape representation. In: 2428
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern 2429
Recognition (CVPR), pp. 165–174 (2019) 2430
2431
2432
2433

[24] Mescheder, L., Oechsle, M., Niemeyer, M., Nowozin, S., Geiger, A.: Occupancy 2434
networks: Learning 3d reconstruction in function space. In: *Proceedings of the* 2435
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2436
2437
2438

2439 pp. 4455–4465 (2019)
2440
2441 [25] Müller, T., Evans, A., Schied, C., Keller, A.: Instant neural graphics primitives
2442 with a multiresolution hash encoding. *ACM Transactions on Graphics* **41**(4), 102
2443 (2022) <https://doi.org/10.1145/3528223.3530127>
2444
2445
2446
2447 [26] Tancik, M., Srinivasan, P.P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N.,
2448 Singhal, U., Ramamoorthi, R., Barron, J.T., Ng, R.: Fourier features let networks
2449 learn high frequency functions in low dimensional domains. In: *Advances in Neural*
2450 *Information Processing Systems*, vol. 33, pp. 7537–7547 (2020). [https://doi.org/](https://doi.org/10.48550/arXiv.2006.10739)
2451 [10.48550/arXiv.2006.10739](https://doi.org/10.48550/arXiv.2006.10739) . arXiv:2006.10739
2452
2453
2454
2455
2456 [27] Maxwell Garnett, J.C.: Colours in metal glasses and in metallic films. *Philo-*
2457 *sophical Transactions of the Royal Society of London A* **203**, 385–420 (1904)
2458 <https://doi.org/10.1098/rsta.1904.0024>
2459
2460
2461
2462 [28] Palik, E.D. (ed.): *Handbook of Optical Constants of Solids*. Academic Press,
2463 Orlando, FL (1985)
2464
2465
2466 [29] Sihvola, A.: *Electromagnetic Mixing Formulas and Applications*. IEE, ??? (1999)
2467
2468
2469 [30] Bruggeman, D.A.G.: Berechnung verschiedener physikalischer konstanten von
2470 heterogenen substanzen. *Annalen der Physik* **416**, 636–664 (1935) [https://doi.](https://doi.org/10.1002/andp.19354160705)
2471 [org/10.1002/andp.19354160705](https://doi.org/10.1002/andp.19354160705)
2472
2473
2474
2475 [31] Hodgkinson, I., Wu, Q.H., Hazel, J.: Empirical equations for the principal refrac-
2476 tive indices and column angle of obliquely deposited films of tantalum oxide,
2477 titanium oxide, and zirconium oxide. *Applied Optics* **37**(13), 2653–2659 (1998)
2478 <https://doi.org/10.1364/AO.37.002653>
2479
2480
2481
2482 [32] Hodgkinson, I., Wu, Q.H.: *Birefringent Thin Films and Polarizing Elements*.
2483
2484

World Scientific, ??? (1997)	2485
	2486
[33] Jamnig, A., Sangiovanni, D.G., Abadias, G., Sarakinos, K.: Atomic-scale diffusion	2487
rates during growth of thin metal films on weakly-interacting substrates. Scientific	2488
Reports 9 , 6640 (2019) https://doi.org/10.1038/s41598-019-43107-8	2489
	2490
	2491
	2492
[34] Poxson, D.J., Mont, F.W., Schubert, M.F., Kim, J.K., Schubert, E.F.: Quantifi-	2493
cation of porosity and deposition rate of nanoporous films grown by oblique-angle	2494
deposition. Applied Physics Letters 93 , 101914 (2008) https://doi.org/10.1063/	2495
1.2981690	2496
	2497
	2498
	2499
	2500
[35] Kesapragada, S.V., Gall, D.: Anisotropic broadening of cu nanorods during glanc-	2501
ing angle deposition. Applied Physics Letters 89 , 203121 (2006) https://doi.org/	2502
10.1063/1.2388861	2503
	2504
	2505
	2506
[36] El Beainou, R.: Relations structure – propriétés de conduction dans des films	2507
W–Cu nanostructurés. Thèse de doctorat, Université de Franche-Comté (Bour-	2508
gogne Franche-Comté) (2019). NNT: 2019UBFCD004. https://theses.hal.science/	2509
tel-02313946	2510
	2511
	2512
	2513
[37] Ben Nacer, S., Jlidi, D., Labidi, A., Chaffar Akkari, F., Touihri, S., Maaref,	2514
M.: Promising ethanol detection enhancement of Cu ₂ O thin film deposited by	2515
GLAD technique. Measurement 151 , 107208 (2020) https://doi.org/10.1016/j.	2516
measurement.2019.107208	2517
	2518
	2519
	2520
	2521
[38] Ben Haj Ali, O., Chaffar Akkari, F., Gallas, B., Martin, N.: Density-Calibrated	2522
Neural Implicit Field for Ballistic Helical GLAD: Scalar Porosity Prediction,	2523
Label-Quality Diagnosis, and the Path to Binary Voxel Occupancy. co-submitted	2524
companion manuscript (2026)	2525
	2526
	2527
	2528
[39] Renaud, G., Lazzari, R., Leroy, F.: Probing surface and interface morphology with	2529
	2530

- 2531 grazing incidence small angle x-ray scattering. Surface Science Reports **64**(8),
2532 255–380 (2009) <https://doi.org/10.1016/j.surfrep.2009.07.002>
2533
2534
2535 [40] Hansen, J.-P., McDonald, I.R.: Theory of Simple Liquids, 4th edn. Academic
2536 Press, ??? (2013)
2537
2538
2539 [41] Chaffar Akkari, F., Trabelsi, I., Abdelkader, D., Gallas, B., Kanzari, M.:
2540 Impedance spectroscopy characterization of anisotropic nano-sculptured copper
2541 oxide Cu₂O thin films for optoelectronic applications. Semiconductor Science and
2542 Technology **34**, 075026 (2019) <https://doi.org/10.1088/1361-6641/ab0ecf>
2543
2544
2545
2546 [42] Álvarez, R., García-Valenzuela, A., Regodón, G., Ferrer, F.J., Rico, V.,
2547 García-Martín, J.M., González-Elipe, A.R., Palmero, A.: Growth dynamics of
2548 nanocolumnar thin films deposited by magnetron sputtering at oblique angles.
2549 Nanotechnology **35**, 095705 (2024) <https://doi.org/10.1088/1361-6528/ad113d>
2550
2551
2552
2553
2554
2555

2556 A Finite-size convergence study

2557 This appendix reports, in full, the box-size (lateral extent L) convergence study
2558 summarised as a limitation in Section 5.4 (Figure 16 shows the full dataset). For
2559 computational tractability this study was carried out at reduced film heights ($h = 25$ –
2560 100 nm, rather than the main study’s $h = 300$ nm) and reports the bulk atom
2561 areal/volumetric density directly (atoms/nm² or atoms/nm³), not the void-fraction
2562 descriptor used elsewhere in this paper; the two are related but not identical observ-
2563 ables, so the numbers below should not be read on the same numerical scale as the
2564 main-text void fractions.

2572 A box-size sweep ($L_x = L_y \in \{75, 100, 125, 150\}$ nm at $\alpha = 85^\circ$, pitch = 150 nm
2573 fixed) shows the mean atom areal density decreases monotonically and substantially
2574 with box size (by $\approx 29\%$ from 75 nm to 150 nm), with no sign of a plateau by $L =$
2575
2576

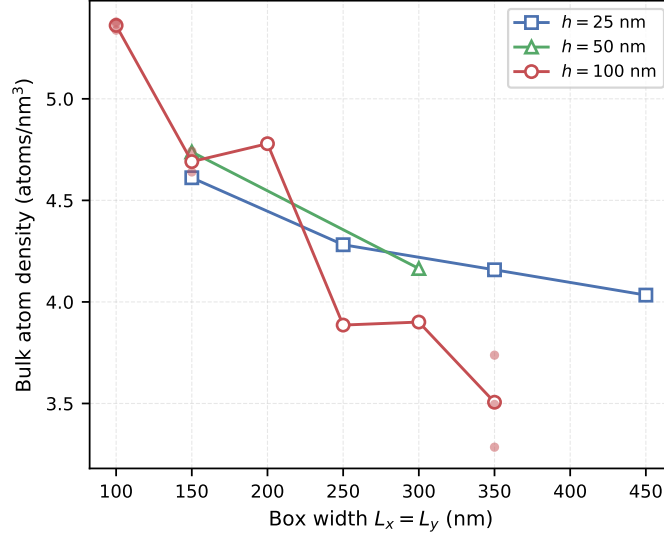


Fig. 16 Bulk atom density falls monotonically with box width at every film height tested, with no sign of a plateau at the box size used in the main study – direct visual evidence for the finite-size limitation discussed in Section 5.4. Bulk atom density versus box width L for all trusted runs in the finite-size convergence campaign, grouped by film height h ($\alpha = 85^\circ$, pitch = 150 nm fixed throughout). Lines connect the mean density at each L ; small dots at $h = 100$ nm, $L = 350$ nm show the individual seed-replicate values behind that mean.

150 nm (Figure 16). The 100 nm box used throughout the main study is therefore *not* at a converged, asymptotic value; the true large-box limit is likely lower still. Separately, a box size exactly equal to the pitch ($L = 150$ nm = pitch) introduces an additional, sharper commensurability artefact (erratic, non-physical atom counts at low α) distinct from the smooth finite-size trend – box sizes equal to (or simple multiples/fractions of) the pitch should be avoided in future campaigns. Attempts to extend the sweep beyond $L = 150$ nm in this first pass were constrained by the simulator’s GPU memory management rather than by physics.

A subsequent height-matched scan ($h = 25$ nm, $L \in \{150, 250, 350, 450\}$ nm) confirmed the density keeps decreasing with no plateau, and showed that a single-variable finite-size scaling collapse (density as a function of $L/(h \tan \alpha)$ alone) does not hold across different heights – at least two physical length scales (the finite-size shadow length and a substrate-proximity transient) govern the box-size dependence,

so the large-box asymptote and a corrected scaling ansatz remain open questions for future work. Cross-checking the $L = 100$ vs 150 nm density change at three angles ($\alpha = 75, 85, 89^\circ$) shows this bias is *not* angle-independent (-32% , -21% , -20% respectively): the angular trend reported in the main text, while qualitatively consistent with the literature comparisons discussed there, should be read as possibly compressed or shifted in magnitude by an angle-dependent finite-size effect that a fixed box size cannot separate from the true physics.

A subsequent degeneracy-breaking campaign (three additional targeted runs: $h = 50$ nm at $L = 150$ and 300 nm, and a second seed at $h = 100$ nm, $L = 350$ nm) extended the height-matched scan above to a third distinct film height and provided the first genuine seed-replicate at large L . Fitting a multiplicative shadow-decay $\times$ substrate-transient model, $\rho(L, h) = \rho_\infty - A(1 - e^{-h/\tau})e^{-L/\lambda}$, with a heteroscedastic weighting exponent p swept over $\{0.5, 0.75, 1.0\}$ (rather than fixed, since p is a closure-model choice, not an estimable constant, at this data volume) gives ρ_∞ point estimates of 4.28 , 4.19 , and 4.03 atoms/nm³ respectively, with 95% bootstrap confidence intervals that overlap across all three p ; the substrate-transient amplitude A is consistently unresolvable (pinned near zero for every p), confirming that only ρ_∞ , not the transient term, is currently informative from this data. Independently, a real-space pair-correlation length $\xi(L)$ measured directly from atom positions (not from the voxelised pipeline) is still increasing at $L = 350$ nm across the $h = 25, 50$, and 100 nm series alike – consistent with, and independent evidence for, the absence of a finite-size plateau reported above. One run in the original height-matched scan ($h = 25$ nm, $L = 250$ nm) had originally failed an automated column-tilt validation gate; this was root-caused to a validator threshold that incorrectly ran a straight-column tilt regression on a genuinely (if only briefly) rotating deposit, producing a spurious result – the underlying atom data for that run is sound and it is retained in the trusted dataset once the validator was corrected.

A resolved but consequential finding: a third seed at $L = 350$ nm was run specifically to place the large- L variance estimate on the same statistical footing ($n = 3$) as the $L = 100$ anchor. With this third replicate, the empirical relative seed-to-seed variance in areal density is confirmed to *grow*, not shrink, with box size: $L = 100$: 0.32% ($n = 3$); $L = 150$: 1.09% ($n = 2$); $L = 350$: 5.28% ($n = 3$) – an approximately $16\times$ increase in relative scatter from the smallest to the largest box, on equal ($n = 3$) statistical footing at the two endpoints, not a two-point artefact. This is the opposite trend from the shrinking variance assumed by every p tested above (all presuppose $N_{\text{col}} \sim L^2$ independent columns self-averaging down the noise), and it is large enough to matter: refitting with a growing-variance weight $\sigma(L) \propto (L/L_{\text{ref}})^q$, with q estimated directly from these three anchors ($q \approx 2.18$) or swept over $\{1.0, 2.0, 2.5\}$, gives $\rho_{\infty} \approx 4.9\text{--}5.2$ atoms/nm³ – materially higher than, and non-overlapping with, the 4.0–4.3 atoms/nm³ range obtained under the originally-assumed shrinking-variance closure. We therefore report ρ_{∞} as 4.9–5.2 atoms/nm³ under the empirically-supported growing-variance weighting, and flag the shrinking-variance value as based on a now-falsified assumption about how simulation noise scales with box size, retained here only for transparency about how much a single modelling choice can move this number. We do not currently have a first-principles explanation for why relative fluctuations grow with L in this simulator; the most plausible hypothesis links it to the non-saturating $\xi(L)$ reported above (a growing transverse correlation length implies the effective number of statistically-independent columns grows more slowly than L^2 , which would suppress rather than enhance self-averaging at large L), but confirming this mechanism – as opposed to merely being consistent with it – is future work. Practically, this means finite-size extrapolation in this system is noise-limited by a growing, not shrinking, per-box uncertainty at large L , so additional seeds at large L (rather than additional, more expensive large- L single runs) are the more statistically efficient way to tighten ρ_{∞} further.

2669
2670
2671
2672
2673
2674
2675
2676
2677
2678
2679
2680
2681
2682
2683
2684
2685
2686
2687
2688
2689
2690
2691
2692
2693
2694
2695
2696
2697
2698
2699
2700
2701
2702
2703
2704
2705
2706
2707
2708
2709
2710
2711
2712
2713
2714

2715 Two claim tiers follow from the box-size result above and should not be conflated.
 2716
 2717 *Quantitative* absolute values reported anywhere in the main text – the void-fraction
 2718 endpoints (88.1%, 98.5%), the PCA variance-explained figure, the CORRNIIF R^2 val-
 2719 ues, and the Maxwell-Garnett birefringence magnitudes – are all computed at the
 2720 100 nm-box convention and are provisional: given the density shifts documented above,
 2721 their numerical values would change at a larger, still-unconverged box size, and none
 2722 should be read as an asymptotic result. *Qualitative*, monotonic angle-orderings (void
 2723 fraction rising with α ; birefringence falling with α in the literature comparison of
 2724 Section 4.5) are the category of claim least exposed to this limitation in principle, but
 2725 at present this project can only offer partial, not conclusive, support for that expec-
 2726 tation: a box-size cross-check exists at just three angles ($\alpha = 75, 85, 89^\circ$), it is itself
 2727 confounded by a simultaneously reduced active-window parameter, and within it the
 2728 openness proxy shifted upward by a small, similarly-sized amount at all three angles
 2729 (consistent with, but not proof of, order-preservation) while a different descriptor com-
 2730 puted on the same box-size campaign, $\beta_{\text{sim}}(\alpha)$, was found to be non-monotonic and its
 2731 box-size dependence remains unresolved. We therefore do not assert that the mono-
 2732 tonic orderings reported here are established to be box-size-robust; they should also
 2733 be treated as provisional pending an angle-resolved convergence study, and are offered
 2734 as the paper’s best-supported trend-level claims rather than as finite-size-independent
 2735 results.
 2736
 2737
 2738
 2739
 2740
 2741
 2742
 2743
 2744
 2745
 2746
 2747
 2748
 2749
 2750
 2751
 2752
 2753
 2754
 2755
 2756
 2757
 2758
 2759
 2760